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Master's Thesis

# Reverberation Mapping Analysis of the 2016 HST Campaign on NGC 4593

prepared by

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# Abstract

NGC 4593 is a nearby Seyfert I Galaxy exhibiting strong variability from the X-ray to the optical bands (McHardy et al. 2017; Cackett et al. 2018) and prominent broad emission-lines (Kollatschny & Dietrich 1997; Cackett et al. 2018; Ochmann et al. 2025). In 2016, the galaxy was monitored with HST/STIS (PI: E. M. Cackett, Program ID 14121) from 12 July to 6 August, spanning wavelength ranges of approximately 1100 Å–1700 Å and 2900 Å–10000 Å.

*Aims:* This thesis aims to perform a reverberation-mapping analysis of the broad-line region (BLR) of NGC 4593 based on the 2016 HST/STIS campaign, to examine its structure, to estimate the mass of the central supermassive black hole (SMBH), and to investigate the variability of the low-ionization line O I $\lambda$8446, which has not previously been monitored in a reverberation-mapping campaign.

*Methods:* Based on the HST/STIS spectra of the 26 epochs, a mean and root-mean-square spectrum were derived. Small wavelength misalignment in the optical spectra between epochs was identified and corrected by intercalibration anchored to the narrow [O III] $\lambda$5007 line. Continua and emission-line light curves were obtained and cross-correlated with the photometric *Swift* UVW2 light curve presented by McHardy et al. 2018 using the interpolated cross-correlation function (ICCF). The time lags between the emission-line and UVW2 light curve were determined by deriving the centroid of the top 20% of the cross-correlation function with uncertainties estimated with FR/RSS Monte Carlo simulations. Line widths were measured taking the FWHM from the rms profiles and the SMBH mass was estimated based on the virial mass of the BLR.

*Results:* This thesis presents the first reverberation-mapping analysis of the BLR in NGC 4593 based on an HST/STIS monitoring campaign. The estimated characteristic BLR radii and line widths of the emission-lines are consistent with an ionization stratification structure of the BLR. Based on the reverberation-mapped lines, an SMBH mass of $\bar{M}_{\rm BH} \approx (5.28 \pm 0.12) \times 10^7\, M_\odot$ was derived, which is in good agreement in order of magnitude with the estimates of earlier studies (e.g. Kollatschny & Dietrich 1997; Denney et al. 2006). In addition, variability of the low-ionization line O I $\lambda$8446 was detected for the first time in a reverberation-mapping campaign. The variability is found to be primarily driven by Ly$\beta$ fluorescence, which allowed the first reverberation-mapping analysis of a Bowen-fluorescence line.

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# 1. Introduction

Active galactic nuclei (AGNs) are among the most luminous objects in the universe, emitting radiation across the entire electromagnetic spectrum (Netzer 2013) and exhibiting strong variability on timescales ranging from hours to days (Ulrich et al. 1997). In contrast to inactive galaxies such as the Milky Way, the supermassive black hole (SMBH) in an active galaxy accretes matter from the central region. This process generates thermal emission from the accretion flow (Peterson 1997) and drives line emission through photoionization and subsequent recombination in the surrounding gas (Netzer 2013). These ionized line-emitting regions are commonly divided into the broad-line region (BLR) and the narrow-line region (NLR) (Peterson 1997). The BLR is located at close distances of light-days to a few light-years from the central SMBH (Goad et al. 2012) and rotates at high velocities of several thousand $\mathrm{km\,s^{-1}}$, which leads to strong Doppler broadening of the emission-lines (Peterson 1997). In contrast, the NLR extends out to several hundred parsecs from the central region (Peterson 1997) with velocities of only a few hundred $\mathrm{km\,s^{-1}}$, resulting in comparatively narrow line profiles.

An interesting example of this class of objects is the barred spiral galaxy NGC 4593. It shows strong variability from the X-ray to the optical bands (McHardy et al. 2017; Cackett et al. 2018) and exhibits several prominent broad emission-lines, including Balmer and helium lines, as well as low-ionization lines like O I $\lambda$8446 (Kollatschny & Dietrich 1997; Cackett et al. 2018; Ochmann et al. 2025). Several studies on NGC 4593 have analyzed the structure and kinematics of its broad-line region (e.g. Kollatschny & Dietrich 1997; Denney et al. 2006; Cackett et al. 2018) using reverberation mapping of the broad emission-lines. This method uses measurable time delays between variations in the ionizing continuum and the response of the line-emitting gas to infer the characteristic radius and geometry of the broad-line region (Peterson 1993).

In a recent study, Cackett et al. 2018 carried out an observation campaign using the Hubble Space Telescope (HST) with the Space Telescope Imaging Spectrograph (STIS) (PI: Cackett, E.M, Prop ID 14121), between 12 July and 6 August 2016,

on NGC 4593 covering wavelength ranges of approximately 1100 Å – 1700 Å and 2900 Å – 10000 Å. This campaign focused on the reverberation of the accretion disk of NGC 4593 by analyzing the UV and optical continua (Cackett et al. 2018). However, a reverberation mapping analysis of the BLR has not been attempted with this dataset yet. As an HST dataset, it is ideally suited for a reverberation mapping analysis, thanks to no atmospheric interference, its daily cadence and wide wavelength coverage. It allows a more detailed structural analysis of the BLR of NGC 4593 compared to earlier campaigns (Kollatschny & Dietrich 1997; Denney et al. 2006; Williams et al. 2018) by including broad emission-lines to the analysis, which have not been monitored for NGC 4593 yet. Building on the HST/STIS dataset presented by Cackett et al. 2018, the first reverberation-mapping analysis of the BLR of NGC 4593 based on an HST/STIS observations is conducted in this thesis.
Furthermore, NGC 4593 shows variability in the low-ionization O I $\lambda$8446 line and the Ca II $\lambda$8498 $\lambda$8542 $\lambda$8662 triplet (Ochmann et al. 2025), which have not been included in reverberation mapping campaigns to this point. Since it is covered within the HST dataset it is therefore possible to include it for the first time in an reverberation mapping analysis. O I $\lambda$8446 is also a known Bowen fluorescence line, which is enhance-able through Ly$\beta$ pumping (Bowen 1947; Grandi 1980). Since the HST dataset covers parts of the UV regime, it is therefore possible to investigate its Bowen fluorescence process for the first time in a reverberation mapping analysis.

# 2. Scientific Background

## 2.1 Active Galactic Nuclei

Active Galactic Nuclei (AGNs) refer to the central region of active galaxies. These objects are among the most luminous in the universe, with bolometric luminosities ranging from $10^{41}$ to $10^{48}$ erg $s^{-1}$ that can outshine entire galaxies by several orders of magnitude (Peterson 1997). Historically, several stellar-based models were proposed for AGNs, such as dense star clusters or supermassive stars. However, these scenarios were discarded, as they are expected to collapse into black holes themselves, and they cannot provide the required energy output. Today, it is understood that the enormous luminosities of AGN are powered by accretion of matter onto a supermassive black hole (SMBH) at the centers of galaxies (Rees 1984). The most widely accepted model for this accretion is a hot, rotating accretion disk surrounding the SMBH, which produces most of the observed radiation (Shakura & Sunyaev 1973). The following sections will outline the key components of AGNs and their variability, which is central to reverberation mapping analysis.

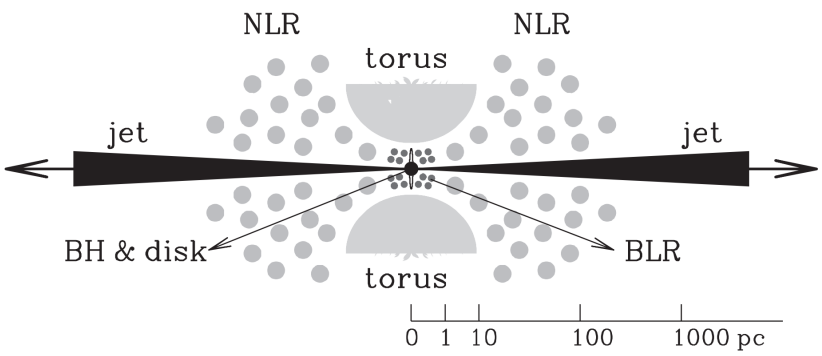


Figure 2.1: Different components of an AGN. Adopted from (Mo et al. 2010). Their Figure 14.3.

### 2.1.1 Geometric and Spectral Features of AGNs

According to the unified model, AGNs consists of a central supermassive black hole (SMBH), a surrounding accretion disk, a dusty torus and ionized gas regions known as the broad-line region (BLR) and narrow-line region (NLR) (Antonucci 1993; illustrated in Figure 2.1). In some cases, relativistic jets are launched perpendicular to the plane of the accretion disc (Urry & Padovani 1995). The following sections describe the physical components of AGN and the spectral features associated with them.

#### Supermassive Black Hole and Accretion Disk

The center of an AGN is defined by a supermassive black hole (SMBH), with typical masses between $10^6\,M_\odot$ and $10^9\,M_\odot$ (Peterson et al. 2004). It does not contribute to the AGN spectrum directly, but acts as the central engine driving observed spectral features of the AGN. It dominates the gravitational potential and, unlike inactive galaxies such as the Milky Way, it is surrounded by an accretion disk. Through viscous processes within the disk, such as turbulent friction and magneto-rotational instability, the angular momentum of the matter is transported outward, which leads to a spiraling inflow of matter toward the SMBH (Shakura & Sunyaev 1973). Several models have been proposed to describe the accretion process. The most widely used model is the geometrically thin and optically thick accretion disk, which consists of ionized gas in differential rotation around the SMBH (Shakura & Sunyaev 1973; Netzer 2013). The disk is composed primarily of ionized hydrogen and helium, with trace amounts of heavier elements (Netzer 2013). The radial extent of the disk is relatively small compared to galactic scales and typically ranges from a few light-hours to a few light-days, corresponding to about $10^{-3}$ to $10^{-2}$ pc (Shakura & Sunyaev 1973; Netzer 2013).

During the accretion process, a substantial fraction of the gravitational energy of the matter is transformed into thermal radiation, which accounts for the enormous luminosity observed in AGNs and heats the accretion disk to high temperatures that depend on the mass of the SMBH (Netzer 2013). For example, the maximum effective temperature for an accretion disk around a SMBH with $M = 10^8, M_\odot$ is on the order of several $\times 10^5$ K, leading to UV and optical emission (Shakura & Sunyaev 1973; Netzer 2013). By contrast, discs around stellar-mass black holes reach much higher temperatures (up to a few $\times 10^6$ K), and therefore emit mostly in X-rays (Shakura & Sunyaev 1973; Netzer 2013). Due to the radial temperature gradient, the emitted spectrum cannot be described as a single blackbody. Instead, it results

from a superposition of many blackbody-like components at different temperatures, often referred to as a multicolour black-body (Netzer 2013). This produces a broad optical–UV continuum of ionizing photons, which interact with gas clouds near the nucleus and play a crucial role in shaping the spectral features of the broad-line and narrow-line region. These photons cause photoionization followed by recombination, which gives rise to the strong emission-lines that are characteristic of AGN spectra (Netzer 2013).

### Broad-Line and Narrow-Line Region

The ionized line-emitting gas clouds can be divided into the broad-line region (BLR) and the narrow-line region (NLR). Both regions differ in density, distance from the SMBH, and the observed line widths (Urry & Padovani 1995). The BLR is located close to the nucleus, at distances ranging from a few light-days to a few light-years from the central SMBH (Goad et al. 2012; see Figure 2.1). It consists of dense gas clouds with electron densities of $n_e \approx 10^{11}\,\mathrm{cm}^{-3}$, moving at velocities of several thousand $\mathrm{km\,s}^{-1}$ due to the strong gravitational influence of the SMBH. These velocities lead to significant Doppler broadening of permitted emission-lines, with widths of (500–10.000) $\mathrm{km\,s}^{-1}$ (Peterson 1997). As described earlier, the BLR is photoionized by the continuum radiation emitted from the accretion disk. Consequently, the line emission from this region responds to changes in the continuum, leading to a strong correlation between the two, and strong variability (Netzer 2013). This relationship is particularly relevant for reverberation mapping, which will be discussed later in Section 2.2.
Modeling the geometry of the BLR is challenging because several emission-lines have to be considered and their intensities vary in response to changes in the continuum radiation (Netzer 2013). A current basic concept locates the BLR just above the accretion disc, between the disc's edge and the dusty torus (Gaskell 2009; Goad et al. 2012), with inflowing and outflowing gas (Czerny & Hryniewicz 2011) and a disc-like distribution (e.g. Williams et al. 2018). It also shows radial ionization stratification (Collin-Souffrin & Lasota 1988; Gaskell & Sparke 1986; Kollatschny et al. 2001), with higher-ionization lines found closer to the SMBH than lower-ionization lines. Typical broad emission-lines include the Balmer, Lyman, and helium lines (Netzer 2013).

The narrow-line region (NLR) extends out to several hundred parsecs from the central region (Peterson 1997). The gas in this region moves at much lower velocities, resulting in emission-lines with widths typically of order (350–400) $\mathrm{km\,s}^{-1}$

(Peterson 1997). In contrast to the BLR, the NLR exhibits both permitted and forbidden transitions. Forbidden lines, such as [O III] $\lambda 5007$, are prominent in the NLR because at its low densities ($n_e \sim 10^2$–$10^4\,\mathrm{cm}^{-3}$) collisional de-excitation is rare, allowing radiative decay from metastable levels (Peterson 1997). Due to its much larger extent compared to the BLR, the NLR responds only very slowly to variations in the ionizing continuum. Therefore, the flux of the narrow emission-lines can be treated as constant over timescales of several years (Peterson 1993).

Because permitted emission-lines can originate in both the BLR and the NLR, they often exhibit a multi-component line profile consisting of a broad and a narrow component (Peterson 1997). The relative strength of the broad and narrow components depends on the fractional contribution of emission from the BLR and the NLR, respectively (Peterson 1997).

#### Dusty Torus

Surrounding the accretion disk and the broad-line region is the dusty torus, a geometrically thick, optically dense structure composed of gas and dust. It extends from the radius at which dust can survive the intense radiation from the accretion disk out to scales of a few parsecs (Netzer 2013). The torus likely has a clumpy distribution and plays a crucial role in the unified model of AGNs (Urry & Padovani 1995; Netzer 2013). The dust in the torus absorbs a significant fraction of the UV and optical radiation emitted by the accretion disk and re-emits it thermally in the infrared. As a result, AGNs typically exhibit strong infrared emission, with the peak wavelength depending on the dust temperature in the torus (Netzer 2013). Even when the central region is hidden from direct view by the torus, this reprocessed infrared emission remains observable. It therefore provides a characteristic signature of obscured AGN activity and enables indirect constraints on the otherwise hidden central engine (Netzer 2013).

### 2.1.2 Classification

AGNs are classified into subgroups based on their spectral features. The key parameters for this classification are luminosity, emission-line profiles and radio properties. Based on those parameters, AGNs get grouped into Seyfert galaxies, quasars and radio galaxies (Peterson 1997). Seyfert galaxies are further subdivided, based on the presence of broad and narrow emission-lines (Peterson 1997), as discussed in the following section. Examples of other subclasses are narrow-line Seyfert 1 galaxies

(NLS1s), low-ionization nuclear emission-line regions (LINERs), and jet-dominated sources such as BL Lac objects or blazars (Antonucci 1993; Urry & Padovani 1995).

### Seyfert Galaxies

Seyfert galaxies are named after Carl K. Seyfert, who in 1943 observed spiral galaxies characterized by exceptionally bright nuclei and strong emission-lines in their optical spectra (Seyfert 1943). They are classified into the sub-classes Seyfert I and Seyfert II based on the presence of broad emission-lines (Peterson 1997), which is demonstrated in Figure 2.2.

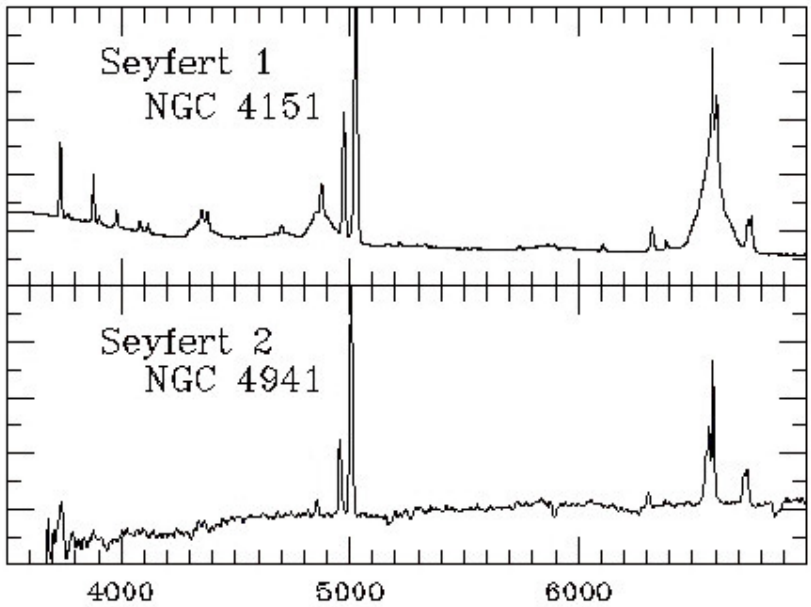


Figure 2.2: A comparison of the spectra of the Seyfert I galaxy NGC 4151 and the Seyfert II Galaxy NGC 4941. While the spectrum of NGC 4151 shows broad and narrow emission-line, the spectrum of NGC 4941 exhibits only narrow emission-lines. Adapted from Keel 2002.

Seyfert I galaxies, show both broad and narrow emission-lines in their optical spectra. Broad lines (e.g the Balmer line H$\alpha$) typically exhibit full widths at half maximum (FWHM) of several thousand kilometers per second and emerge from the fast-moving, high-density gas in the BLR (Peterson 1997). In contrast, narrow lines, including prominent forbidden transitions like [O III] $\lambda$5007 or [N II] $\lambda$6584, originate from the slow-moving, low-density gas in the NLR (Peterson 1997). An AGN showing both components in its spectrum is classified as a Seyfert I galaxy (Peterson 1997), which is the case for NGC 4593. Further details on NGC 4593 are provided in Section 3.1. Between the two main Seyfert classes, several intermediate subclasses (1.2, 1.5, 1.8, 1.9) are defined based on the relative strength of the broad

and narrow Balmer-line components in the optical spectrum (Osterbrock 1977; Osterbrock 1981; Peterson 1997). Seyfert 1.8 and 1.9 galaxies show very weak broad components. In Seyfert 1.9 objects, the broad component is visible only in the H$\alpha$ line, whereas in Seyfert 1.8 objects it is also detectable in H$\beta$. Furthermore, if the broad and narrow components in H$\beta$ are of equal strength, the AGN is classified as a Seyfert 1.5 (Peterson 1997). If the narrow component is even weaker than the broad component, it is classified as a Seyfert 1.2 (Osterbrock 1977).

Seyfert II galaxies lack broad-line emission in their optical spectra, most likely due to orientation-dependent obscuration by a dusty torus (Antonucci 1993; Peterson 1997).This implies that the classification of Seyfert galaxies depends strongly on the observer's viewing angle, which is a central element of the unified model of AGN (Antonucci 1993; Peterson 1997). The unified model is discussed in more detail in Section 2.1.3.
Another notable subclass of Seyfert galaxies is narrow-line Seyfert I galaxies (NLS1s). They are commonly defined by the comparatively small width of the typically broad H$\beta$ emission-line (FWHM $< 2000\,\mathrm{km\,s^{-1}}$) and weak [O,III] $\lambda$5007 emission relative to H$\beta$ (Osterbrock & Pogge 1985; Komossa 2007), and they typically show strong Fe II emission (Véron-Cetty et al. 2001). Strong Fe II emission often correlates with weak [O III] emission and small Balmer-line widths, which provides strong constraints on the physical properties of the central region of an AGN and makes NLS1s of particular interest for study (Osterbrock & Pogge 1985; Komossa 2007).

### Additional AGN Classes

In addition to Seyfert galaxies, there are several other classes of AGNs. Quasars (quasi-stellar objects) are more luminous than Seyfert galaxies, typically outshine their host galaxies, and are usually observed at higher redshifts (Netzer 2013). Because quasars show emission properties identical to those of Seyfert galaxies, the modern distinction is primarily based on luminosity: quasars are high-luminosity AGNs, whereas Seyfert galaxies occupy the lower-luminosity end (Netzer 2013).
Radio galaxies form another important AGN class, distinguished by their strong radio emission and prominent jets and are typically found in elliptical host galaxies (Netzer 2013). When their jets are aligned close to the observers line of sight, they are observed as blazars or BL Lac objects, which exhibit rapid variability and featureless optical spectra due to relativistic beaming (Netzer 2013).
Finally, LINERs are low-luminosity AGNs with spectra dominated by low-ionization emission-lines. The physical origin of their ionization mechanism is still debated,

and in some cases, they may not be powered by accretion at all (Netzer 2013).
While these classifications are based primarily on spectral characteristics, many of the observed differences between AGN types can be attributed to orientation effects. The Unified Model of AGN provides a framework that explains this apparent diversity through a common internal structure, viewed from different angles.

### 2.1.3 Unification Model

Figure 2.3 illustrates the unified model of AGN proposed by Antonucci 1993. He argued that the observed differences among AGN spectra do not require fundamentally different intrinsic structures. Instead, they arise mainly from orientation of the central region towards the observer and from its obscuration by the dusty torus.
The figure shows how the same AGN would be classified depending on the observer's viewing angle. As mentioned above, the dusty torus plays a key role, as it surrounds the SMBH, the accretion disk and the fast-moving BLR. If the line of sight is blocked by the torus, the broad-line emission is obscured, resulting in a Seyfert II spectrum, exhibiting only narrow emission-lines. In this geometry, the AGN is viewed at high inclination, with the torus intercepting the line of sight. If, on the other hand, the nucleus is viewed directly and is not obscured by the torus, both broad and narrow emission-lines are observed in the spectrum, and the AGN is classified as a Seyfert I galaxy (Urry & Padovani 1995).
The same principle applies to other AGN classes. Quasars can be considered the high-luminosity counterparts of Seyfert galaxies, where orientation and torus obscuration likewise affect their observed properties. Blazars, are seen when the relativistic jet is aligned closely with the observer's line of sight, leading to strong Doppler boosting, which makes the radiation appear significantly brighter and shifted to higher frequencies than it intrinsically is (Urry & Padovani 1995).
Although the classical Unification Model treats AGN classification as fixed and purely geometry-driven, some AGNs have been observed to change their spectral type over time (e.g. Ricci & Trakhtenbrot 2023). These so-called "changing-look AGNs" demonstrate that a purely orientation-based interpretation, such as the Unification Model, cannot explain all observed phenomena. They suggest that intrinsic changes, such as variations in accretion rate or obscuring material, can also affect the classification (Ricci & Trakhtenbrot 2023).

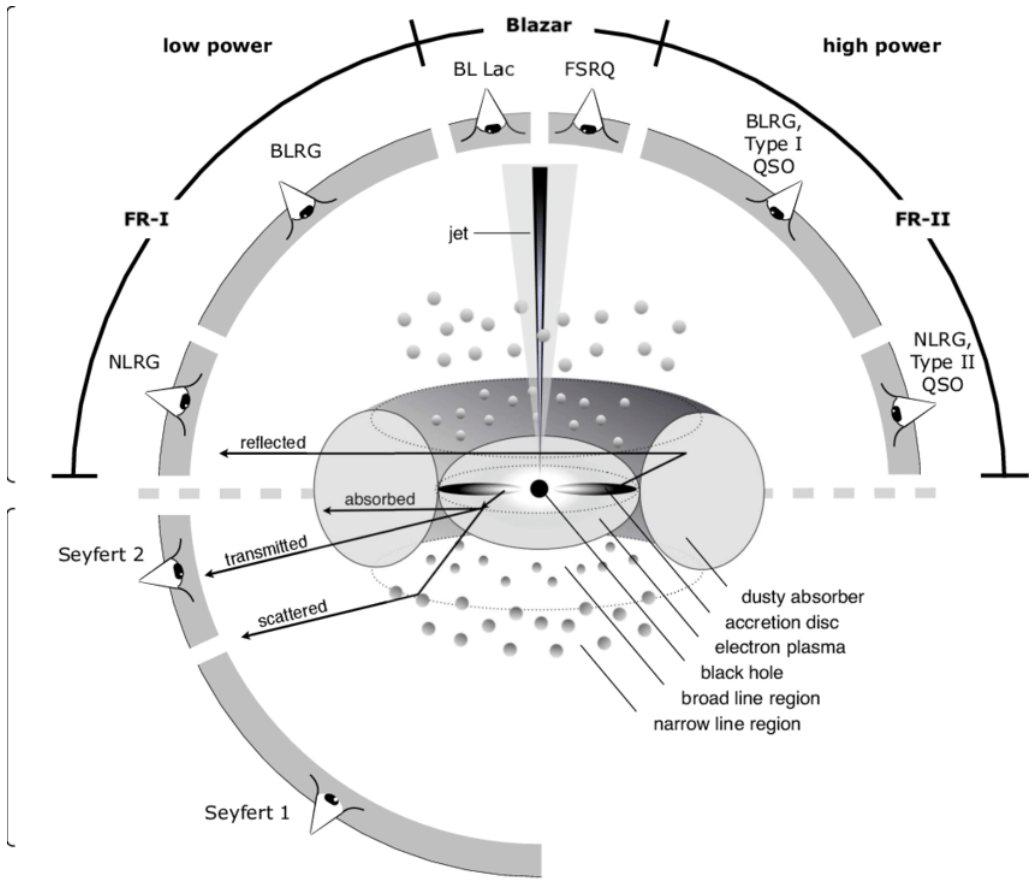


Figure 2.3: This graphic shows a schematic of the unification model of an AGN. The figure was adopted from Beckmann & Shrader 2013.

### 2.1.4 Variability

The variability of active galactic nuclei is one of the key aspects that enables the study of their central regions, which generally cannot be probed with conventional spatially resolved observations. Variability is observed on timescales ranging from hours to several years and is generally stochastic, resulting in flux variations of both emission-lines and continuum emission of up to a few tens of percent in the UV and optical bands (Ulrich et al. 1997). Although the origin of this variability is not yet fully understood, the most widely accepted models attribute it to inhomogeneities and instabilities within the accretion disk (Ulrich et al. 1997; Dexter & Agol 2010). Depending on the underlying physical process, variations occur on different characteristic timescales. Processes such as thermal fluctuations or changes in the accretion flow happen on timescales of decades to centuries for typical SMBH masses and radii, and are therefore difficult to observe directly. In contrast, processes operating on shorter timescales are easier to study. Examples include gas motions and mechanical instabilities (e.g., sound waves) within the disk, which occur on timescales of weeks

to months (Ricci & Trakhtenbrot 2023). The shortest timescale is the light-crossing timescale, $t_{\rm lc} = R/c$, which specifies how long light takes to traverse the emitting region (e.g. BLR) (Ricci & Trakhtenbrot 2023). Here, $c$ denotes the speed of light, and $R$ denotes the characteristic size or radius of the variable emitting region. Following Ricci & Trakhtenbrot 2023, assuming a SMBH of mass $\approx 10^8\, M_\odot$, the light-crossing timescale can be written as

$$t_{\rm lc} = \frac{R}{c} \simeq 0.87 \left(\frac{R}{150\, r_g}\right) M_8 \text{ days}, \tag{2.1}$$

where $r_g = GM/c^2$ is the gravitational radius of the black hole. Thus, the light-crossing timescale of the variable emitting region is of order days, and $t_{\rm lc}$ scales linearly with the size of the emitting region (Ricci & Trakhtenbrot 2023).
Because variations in the ionizing continuum occur on such short timescales, it is possible to measure delayed responses from other regions within the AGN that are correlated with the continuum, using long-term monitoring campaigns (Peterson 1997). In particular, the BLR responds to changes in the photoionizing continuum radiation of the central source with a time lag that is longer than the light-crossing timescale of the emitting region (Peterson 1997). This lag forms the basis of classical reverberation mapping, which will be further elaborated in the next section.

## 2.2 Reverberation Mapping

The main focus of this work is a classical reverberation mapping analysis of the broad lines of NGC 4593. Reverberation mapping probes the structure of the BLR around the SMBH by measuring the time lag between continuum variations and the correlated response of the broad lines. This lag can be used to constrain the BLR geometry and to estimate the SMBH mass (Peterson 1993).

### 2.2.1 Principle of Reverberation Mapping

The fundamental assumption in reverberation mapping is that variations in the observed continuum flux are echoed by variations in the emission-line flux, with a measurable time lag. When the continuum luminosity varies, the emission-line response follows with a measurable time lag (Peterson 1993). The time lag $\tau$ corresponds to the average light-travel time between the photo-ionizing continuum source and the line-emitting regions. Following the derivation of Peterson 1997, $\tau$ can be written as

$$\tau = (1 + \cos\theta) \cdot \frac{R_{\rm BLR}}{c}, \tag{2.2}$$

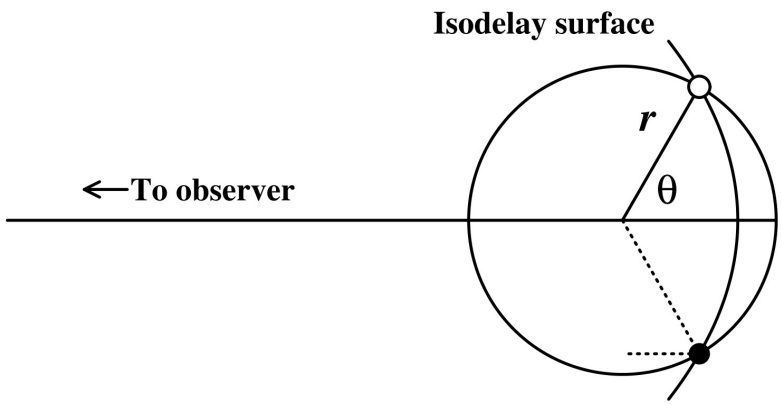


Figure 2.4: Isodelay surface, adopted from (Peterson & Horne 2004).

where $R_{\mathrm{BLR}}$ is the characteristic BLR radius, $c$ is the speed of light, and $\theta$ is the angle between the line of sight and the position vector of the emitting gas with respect to the central source (see Figure 2.4; Peterson 1997). For a fixed lag $\tau$, the emitting regions that respond at that delay lie on a paraboloid aligned with the observer's line of sight, known as an isodelay surface, represented by the circle in Figure 2.4. Therefore, emission observed at a given lag $\tau$ originates from the intersection of the BLR distribution with the corresponding isodelay surface (Peterson 1997). Thus, reverberation mapping can be used to "map" the BLR by inferring a characteristic BLR radius from the measured time lag (Peterson 1997). However, the observer receives emission from a range of delays (i.e., effectively from many isodelay surfaces), so the so-called transfer equation is required, which integrates over all delays (Peterson 1997):

$$L\left(t\right) = \int \Psi\left(\tau\right) C\left(t-\tau\right) d\tau. \tag{2.3}$$

Here, $\Psi(\tau)$ is the transfer function, which encodes the BLR geometry and kinematics, $C(t)$ is the continuum light curve, and $L(t)$ is the emission-line light curve (Peterson 1997). Although the BLR response is, in principle, fully described by the transfer function $\Psi(\tau)$, the lag is commonly estimated using cross-correlation techniques. In practice, recovering the full transfer function $\Psi\left(\tau\right)$ requires densely sampled, high signal-to-noise light curves spanning a duration much longer than the expected lag. Since real monitoring campaigns are often affected by observational gaps and noise, such reconstructions are rarely possible (Horne et al. 2004; Peterson 1993). For this reason, this project focuses on measuring the mean time lag between continuum and emission-line variations using the interpolated cross-correlation function (ICCF) method (Gaskell & Peterson 1987), which is well established in reverberation mapping campaigns (e.g. Kollatschny & Dietrich 1997; Denney et al. 2006; Williams et al. 2018; Probst et al. 2025).

### 2.2.2 Lag Measurement

Using the notation of Peterson 1997, the cross-correlation function between the ionizing continuum and an emission-line is defined as

$$F_{\mathrm{CCF}}(\tau) = \int_{-\infty}^{\infty} L(t)C(t-\tau)dt. \tag{2.4}$$

The auto-correlation function of the ionizing continuum is

$$F_{\mathrm{ACF}}(\tau) = \int_{-\infty}^{\infty} C(t)C(t-\tau)dt. \tag{2.5}$$

Together with the transfer equation (Equation 2.3), the cross-correlation function can be written as the convolution of the transfer function and the auto-correlation function of the ionizing continuum:

$$F_{\mathrm{CCF}}(\tau) = \int_{-\infty}^{\infty} \Psi(\tau')F_{\mathrm{ACF}}(\tau-\tau')d\tau'. \tag{2.6}$$

The lag is commonly defined as either the peak location ($\tau_{\mathrm{peak}}$) or the centroid ($\tau_{\mathrm{centroid}}$) of the CCF (Peterson 1997). While $\tau_{\mathrm{peak}}$ is defined as the location of the CCF maximum, $\tau_{\mathrm{centroid}}$ is calculated over all CCF points above a selected threshold, typically 80% of the peak value. Because the CCF is closely related to the transfer function, it is possible to infer a characteristic BLR size associated with the emission-line response (Peterson 1997), which can be expressed as

$$R_{\mathrm{BLR}} = c \cdot \tau_{\mathrm{centroid}}. \tag{2.7}$$

This follows from the light-travel timescale (Peterson et al. 2004). Since the centroid lag is generally considered a more robust estimator of the mean BLR light-travel time of the BLR (Peterson et al. 2004), it is used in this project.
The uncertainty of the measured lag is estimated using a Monte Carlo approach combining flux randomization (FR) and random subset selection (RSS) (Peterson et al. 1998; Peterson et al. 2004). In the FR step, each flux value is randomly perturbed according to its measurement uncertainty. In the RSS step, $N$ data points are drawn randomly with replacement, while duplicate selections are discarded, resulting in a new light curve with $M \leq N$ points. For each realization, the ICCF analysis is repeated, yielding a distribution of centroid lags. The uncertainties are estimated from the distribution of centroid lags obtained from the simulations (Peterson et al.

2004). The 16th and 84th percentiles of this distribution are adopted as the bounds of the $1\sigma$ confidence interval (Peterson et al. 1998).

### 2.2.3 Black Hole Mass

The reverberation mapping method can be used not only to measure the characteristic size of the BLR, but also to estimate the mass of the central SMBH. Under the assumption that the gas dynamics in the BLR are dominated by the gravitational potential of the central SMBH, the black hole mass can be estimated using the virial theorem (Peterson et al. 2004).
The centroid time lag $\tau_{\text{centroid}}$ provides an estimate of the characteristic BLR radius via Equation 2.7. Together with the velocity dispersion $\Delta V$ of the BLR gas, the virial mass is given by

$$M_{\text{vir}} = \frac{R_{\text{BLR}}\,\Delta V^2}{G}. \tag{2.8}$$

The black hole mass is then given by

$$M_{\text{BH}} = f \cdot M_{\text{vir}}. \tag{2.9}$$

Here, $G$ denotes the gravitational constant, and $f$ is a scale factor that accounts for the unknown geometry, kinematics, and inclination of the BLR (Peterson et al. 2004). The velocity dispersion $\Delta V$ can be estimated from the widths of the broad emission-lines (Peterson et al. 2004).
The scale factor $f$ is calibrated by matching reverberation-based black hole masses to the empirical $M_{\text{BH}} - \sigma_*$ relation observed in inactive galaxies, where $\sigma_*$ denotes the stellar velocity dispersion of the galactic bulge (Onken et al. 2004). Different studies have reported values of $f$ based on various AGN samples, for example, $f = 5.5$ (Onken et al. 2004), $f = 4.31$ (Grier et al. 2013), and $f = 3.6$ (Graham et al. 2011). The calibration of the scale factor also depends on the measurement method used for the line width of the broad emission-lines (Peterson et al. 2004). Two commonly used measures are the line dispersion $\sigma_{\text{line}}$ (see Peterson et al. 2004) and the FWHM. The previously named scale factories where estimated with the line dispersion measure. Since the FWHM measure is used in this work, a average scale factor of $f = 1.8$ gets adopted, following Probst et al. 2025. This value bases on the average scale factor $f = 3.6$ reported by Graham et al. 2011 and takes the relation $\text{FWHM}/\sigma_{\text{line}} \approx 2$ Peterson et al. 2004 into account.

## 2.3 Bowen Fluorescence of OI$\lambda$8446

Previous studies of NGC 4593 report strong emission in the low-ionization lines O I $\lambda$8446 and the Ca II $\lambda$8498 $\lambda$8542 $\lambda$8662 triplet (Ochmann et al. 2025). In particular, O I $\lambda$8446 is of interest because it can be enhanced through a fluorescence mechanism referred to as Bowen fluorescence (Bowen 1947; Grandi 1980). The mechanism was first described by I.S. Bowen in 1934 to explain unexpected emission-lines in nebular spectra. This mechanism describes a resonant line-pumping process in which photons emitted by one ion are absorbed by another ion of a different species via a permitted transition enabled by a near coincidence in wavelength between the pumping line and the absorbing transition. The resulting de-excitation leads to an enhancement of the emission-lines (Bowen 1934).

One transition that can be enhanced by Bowen fluorescence is O I $\lambda$8446, which can be pumped by Ly$\beta$ photons (Bowen 1947; Netzer & Penston 1976). In this process, Ly$\beta$ photons at $\lambda$1025.72 Å are absorbed by neutral oxygen through the near-resonant transition $2p^4\,{}^3P_2 \rightarrow 3d\,{}^3D^0$ of O I at $\lambda$1025.77 Å. The excited $3d\,{}^3D^0$ level decays to $3p\,{}^3P$, which then decays to $3s\,{}^3S^\circ$, emitting the O I $\lambda$8446 emission-line (see Figure 2.5; Grandi 1980). This provides an additional excitation channel for O I $\lambda$8446, in addition to recombination, continuum fluorescence or collisional excitation.

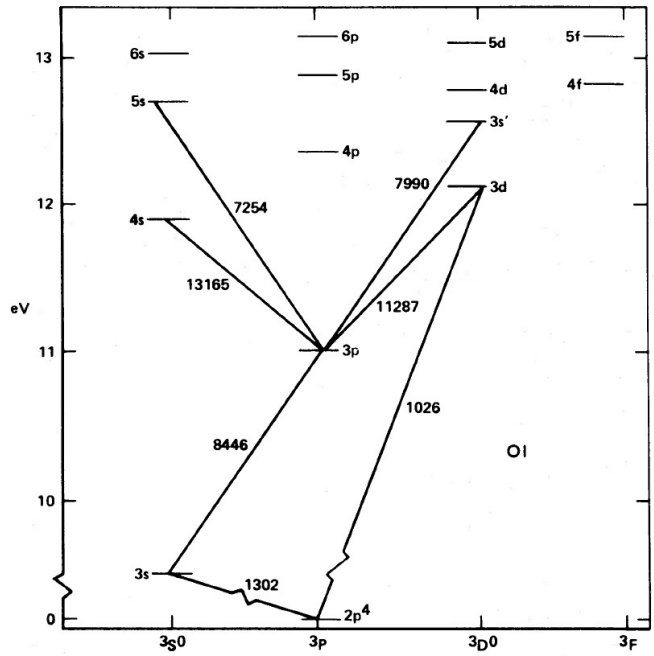


Figure 2.5: Energy level diagram displaying the process of Bowen fluorescence pumping of OI, adopted from (Grandi 1980).

# 3. Campaign and Data Preparation

The analysis of this campaign is based on the HST dataset of the observation campaign of NGC 4503 in 2016 (PI: Cackett, E.M, Prop ID 14121). This campaign took place between the 12th of July and the 6th of August with daily observations, which resulted in 26 successful out of 27 observations. It was performed with the Hubble Space Telescope (HST) using the Space Telescope Imaging Spectrograph (STIS) with the three different gratings. The following section will cover an overview of the properties and specifications of NGC4593 and the campaign in 2016.

## 3.1 NGC4593

NGC 4593 is classified as a Seyfert 1 galaxy with a barred spiral morphology of type (R)SB(rs)b (Denney et al. 2006). It is located at RA = $12^{\mathrm{h}}39^{\mathrm{m}}39.44^{\mathrm{s}}$, Dec = $-05°20'39.03''$ (J2000) and has a redshift of $z = 0.0083 \pm 0.0005$, corresponding to a distance of $\sim 35.9\,\mathrm{Mpc}$ (NED 2025) assuming a $\Lambda$CDM cosmology. The galaxy exhibits a prominent large-scale bar and nuclear dust ring connected to dust lanes along the bar, as seen in Figure 3.1.

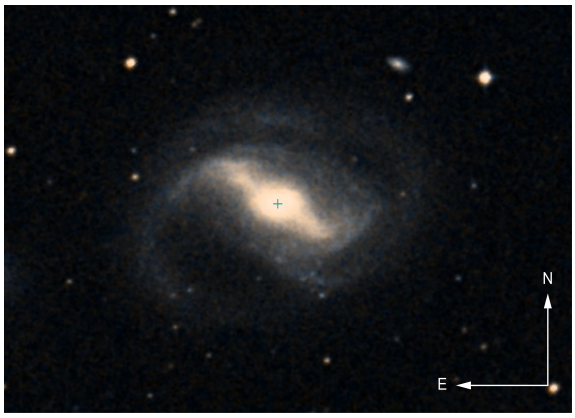


Figure 3.1: Screenshot of NGC 4593 visualized with Aladin Lite (*Aladin Lite* 2025) using DSS2 survey imagery (STScI 2025). The image is oriented with north up and east to the left. Right ascension increases to the left and declination increases upward.

The AGN in NGC 4593 exhibits strong broad emission lines, including Balmer and Lyman lines as well as He, O, and Ca lines among others (Kollatschny & Dietrich 1997; Denney et al. 2006; Cackett et al. 2018; Ochmann et al. 2025). Several variability and reverberation-mapping campaigns have monitored different broad emission lines(e.g. Dietrich et al. 1994; Kollatschny & Dietrich 1997; Denney et al. 2006. They reported FWHM values from the mean (AVG) and root-mean-square (RMS) spectra of their respective campaigns. The RMS spectrum is defined as the standard deviation of the flux at each wavelength across epochs:

$$F_{\mathrm{RMS}}(\lambda) = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}\left[F_i(\lambda) - \bar{F}(\lambda)\right]^2}\,, \tag{3.1}$$

with the mean spectrum at wavelength $\lambda$ given by

$$\bar{F}(\lambda) = \frac{1}{N}\sum_{i=1}^{N} F_i(\lambda)\,. \tag{3.2}$$

Kollatschny & Dietrich 1997 measured $\mathrm{FWHM}_{\mathrm{AVG/RMS}} = (3400 \pm 200)\,\mathrm{km\,s^{-1}}$ for H$\alpha$, while Denney et al. 2006 reported for H$\beta$ $\mathrm{FWHM}_{\mathrm{AVG}} = (5142 \pm 16)\,\mathrm{km\,s^{-1}}$ and $\mathrm{FWHM}_{\mathrm{RMS}} = (4141 \pm 416)\,\mathrm{km\,s^{-1}}$. Based on these broad-line widths, they estimated the SMBH mass to be $M \approx 1.4 \times 10^7\,M_\odot$ (Kollatschny & Dietrich 1997) and $M = (9.8 \pm 2.1) \times 10^6\,M_\odot$ (Denney et al. 2006). Overall, these results suggest that the SMBH mass is of order $10^7\,M_\odot$.
Furthermore, NGC 4593 shows a rare double-peaked emission-line complex involving O I $\lambda$8446 and the Ca II $\lambda$8498, $\lambda$8542, $\lambda$8662 triplet (Ochmann et al. 2025). Ochmann et al. 2025 found that the Ca II triplet has an intensity ratio of 1:1:1 and line profiles that closely resemble those of O I $\lambda$8446, exhibiting a red-to-blue peak ratio of 4:3 and a FWHM of $\approx 3700\,\mathrm{km\,s^{-1}}$, suggesting that these lines originate in a similar high-density emission region. A fit of the asymmetric line profile of Ca II $\lambda$8662 with an elliptic accretion disk model, shows an eccentric of $e \approx 0.22$ and a low-inclination of $i \approx 11°$.

## 3.2 The 2016 HST Campaign

The HST campaign (PI: Cackett, E.M, Prop ID 14121) was designed to study wavelength-dependent continuum lags. It took place between the 12th of July and the 6th of August with daily observations, which resulted in 26 successful out of 27 observations with daily cadence. Observations were carried out with the Hubble Space Telescope (HST) using the Space Telescope Imaging Spectrograph (STIS)

and three different gratings. The low-resolution STIS gratings provided continuous spectral coverage over a broad wavelength range of approximately 1100 Å – 1700 Å and 2900 Å – 10000 Å. In each observation, spectra were obtained using the G140L, G430L, and G750L. All spectra were acquired with the $52'' \times 0.2''$ slit.
The characteristics of the STIS gratings used in this work are summarized in Table 3.1. After standard pipeline processing, charge-transfer inefficiency (CTI) corrections were applied using an algorithm based on (Anderson & Bedin 2010). The remaining hot pixels were removed manually by M. Ochmann.
The 27 spectra are retrieved from the *Hubble Advanced Spectral Products (HASP)* 2025 archive using the HASP search form in *Mikulski Archive for Space Telescopes (MAST)* 2025. 26 of these spectra are usable for further analysis. The top panel of Figure 3.2 shows all spectra in the spectral range from 4000Å to 9000Å.

Table 3.1: Overview of STIS grating characteristics (STIS 2025).

| **Grating** | **Range [Å]** | **Exp. Time [s]** | **Res. Power** | **Dispersion [Å/pixel]** |
|---|---|---|---|---|
| G140L | 1119–1715 | 1234 | $\sim 1000$ | 0.6 |
| G430L | 2888–5697 | 298 | $\sim 500$ | 2.73 |
| G750L | 5245–10233 | 288 | $\sim 500$ | 4.92 |

## 3.3 Intercalibration and Determination of mean and rms Spectra

For the subsequent analysis, the mean spectrum over all epochs is obtained, improving the signal-to-noise ratio. Furthermore it is essential for the reverberation mapping analysis to identify variability between the epochs, which can be assessed with the root-mean-square (rms) spectrum, defined as the standard deviation of the flux at each wavelength across epochs (see. Equation 3.1). Constant features, such as narrow emission lines, vanish in the RMS spectrum, leaving only variable components such as broad emission lines. The top panel of Figure 3.2 shows the mean and rms spectra from the original retrieved data. Residual variability is still noticeable in nominally non-varying lines, particularly in the forbidden line [O III] $\lambda$5007. This indicates small wavelength misalignment between epochs. Therefore, an intercalibration anchored to the narrow [O III] $\lambda$5007 line was performed, which is common practice in RM studies (e.g. Dietrich et al. 1994; Denney et al. 2006). This was achieved by shifting the wavelengths of the individual spectra and scaling the [O III] $\lambda$5007 line flux to a constant value. As a narrow line, the flux of [O III] $\lambda$5007 can

be assumed to remain constant over the timescale of the campaign. Based on this assumption, the [O III] flux in each spectrum was scaled to $(106\pm5)\times10^{-15}\,\mathrm{erg\,s^{-1}\,cm^{-2}}$ and the wavelength was shifted by a maximum of 1Å.
Figure 3.2 shows a comparison of the original and the intercalibrated epochs and the corresponding mean and rms spectra. The disappearance of narrow features in the calibrated rms spectrum, especially the [O III] $\lambda$5007 line, confirms that the apparent variability in the rms of the original epochs was induced by the wavelength shifts between them, rather than intrinsic line variability. However the intercalibration was only applied to the optical regime of the spectra, because a narrow-line flux calibration is only valid over a limited wavelength regime. In the following analysis, the intercalibrated mean and rms spectra are used for the optical range obtained with the G430L and G750L gratings, while the mean and rms spectra from the original epochs are used for the UV emission-line analysis, as these were acquired with the G140L grating.

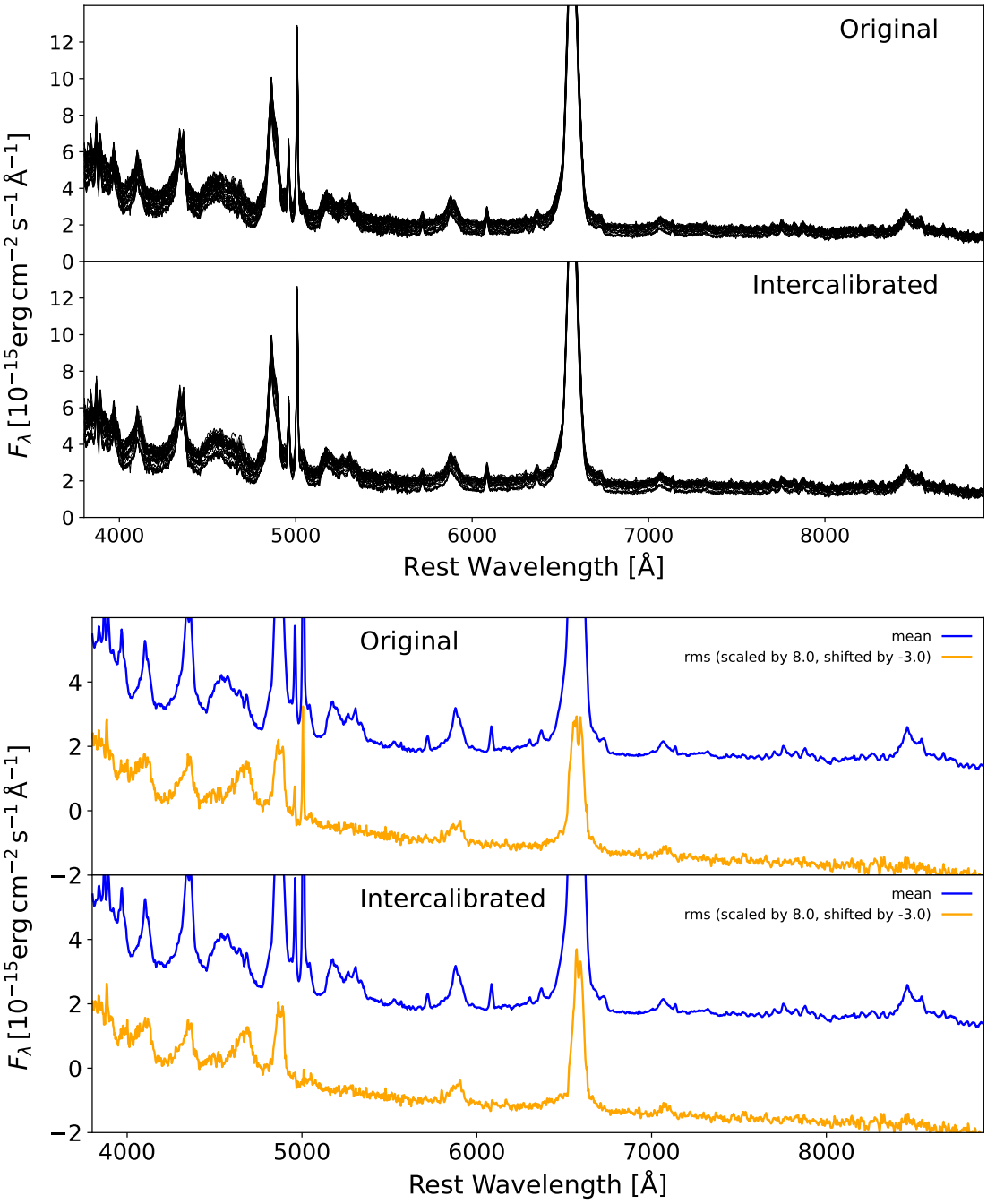


Figure 3.2: Comparison of the spectral range between 4000Å and 9000Å from the 2016 HST campaign of NGC 4593, showing the effects of [O III] $\lambda$5007 intercalibration on both the individual spectra (top) and the derived average and rms spectra (bottom).

# 4. Reverberation Mapping Analysis of NGC4593

## 4.1 Line Identification

The reverberation mapping analysis begins with line identification in the derived mean spectrum of the HST/STIS dataset. This is done for the optical–NIR range 3900 Å–9000 Å and for the UV range 1100 Å– 1700 Å.
In the optical–NIR range (see Figure 4.1), several prominent broad emission-lines are identified: Balmer lines from H$\alpha$ – H$\epsilon$; He I $\lambda$4471, He I $\lambda$5016, He I $\lambda$5876 and He I $\lambda$7065; He II $\lambda$4686; and the low-ionization lines O I $\lambda$8446 and the Ca II $\lambda$8498 $\lambda$8542 $\lambda$8662 triplet. All of these lines show some level of variability in the rms spectrum. For the subsequent analysis, the Balmer lines H$\alpha$ – H$\delta$, He II $\lambda$4686 and He I $\lambda$5876 are selected, as they show the most pronounced variation in the rms spectrum. The low-ionization line O I $\lambda$8446 shows noticeable variability, which was not monitored in an reverberation mapping analysis yet and is therefore included as well. Two prominent Fe II emission-line bands are also present in the optical spectrum as well, one between $\sim$ 4489 Å – 4629 Å, blending with He I $\lambda$4471 and He II $\lambda$4686, and another between $\sim$ 5169 Å – 5336 Å. The mean spectrum also shows several narrow forbidden emission-lines, with [O III] $\lambda$4363, $\lambda$4959, and $\lambda$5007 being the most prominent. As they are assumed to be constant over the timescale of the campaign, they show no significant variability in the rms spectrum.
In the UV spectrum (Figure 4.2), five broad emission-lines are identified: Ly$\alpha$, which overlaps with the N V $\lambda\lambda$1238, 1242 doublet; Si IV $\lambda\lambda$1393, 1402; C IV $\lambda\lambda$1548, 1550; and He II $\lambda$1640. Since the UV emission-lines of NGC 4593 were not monitored in an reverberation mapping analysis yet, these emission-lines will be included as well in the subsequent analysis. In addition to the broad emission-lines, the UV spectrum exhibits semi-forbidden emission-lines, like the O III] $\lambda\lambda$ 1660, 1666 doublet, which partially blends with He II $\lambda$1640.

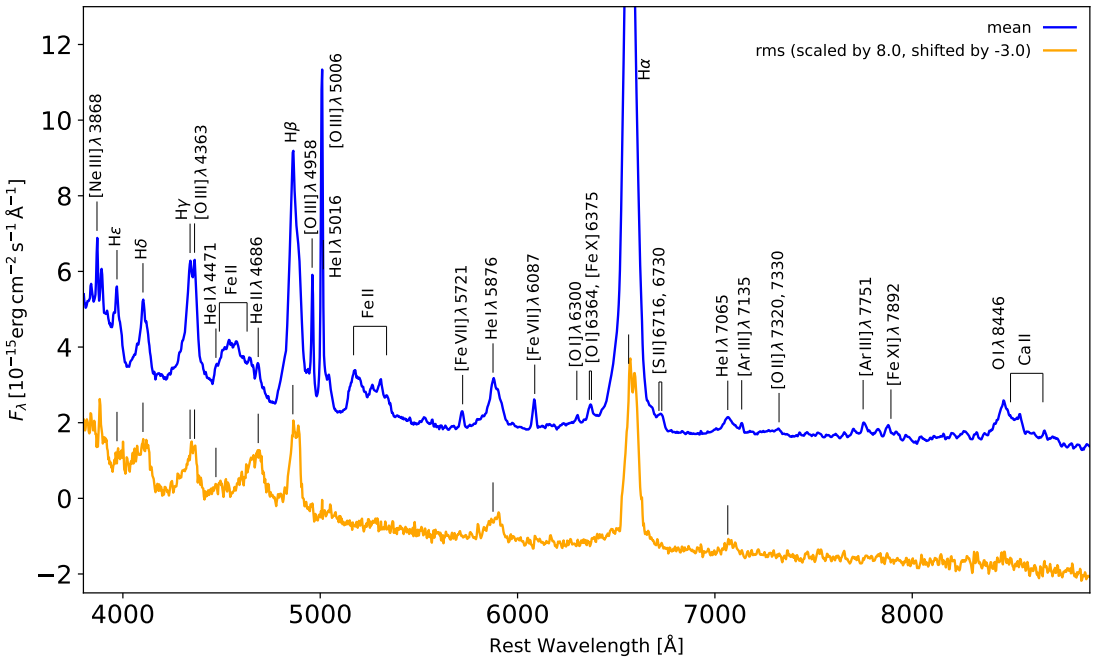


Figure 4.1: Optical-to-NIR mean and rms spectrum with identified emission-lines.

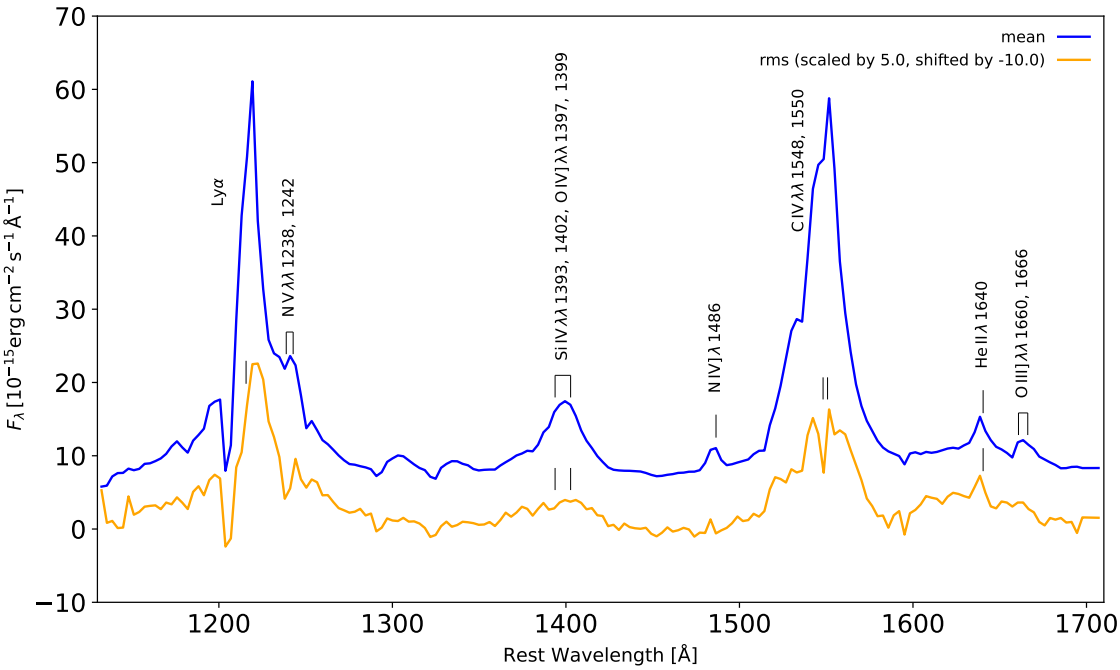


Figure 4.2: UV spectrum mean and rms spectrum with identified emission-lines.

## 4.2 Emission-Line and Continua Measurement

After the identification and selection of the broad emission-lines, their fluxes are measured in every epoch to extract their light curves. This is performed with a Python-based tool called GECHO, developed by M. Probst. This tool is able to import full campaigns, determine mean and rms spectra, extracting light curves and conduct further measurements, naming here line-width measurements and lags measurement based on the methods of Gaskell & Peterson 1987 and Peterson et al. 2004, which are further discussed in Sections 4.5 and 4.4.
The emission-line light curve extraction follows the same principle of previous reverberation mapping campaigns (e.g. Kollatschny & Dietrich 1997; Probst et al. 2025). Figure 4.3 displays the GECHO graphical user interface (GUI), which provides a side-by-side view of the campaign's mean and rms spectra. The line flux at each epoch is obtained by integrating the flux density over the wavelength range, marked in red in Figure 4.3. The integration boundaries are chosen to include the variable part of the emission-line while excluding contributions from neighboring lines. To account for the surrounding continuum, a linear pseudo-continuum is interpolated from two line-free wavelength windows on the blue and red side of the emission-line, shown in grey in Figure 4.3. The adopted integration limits and pseudo-continua for each line are listed in Table 4.1.

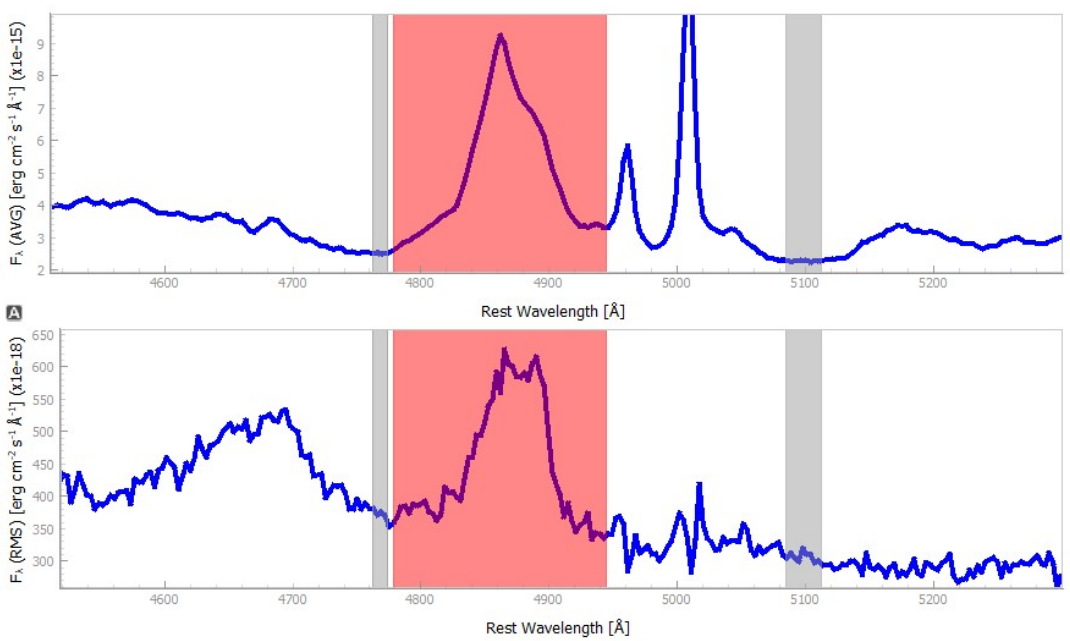


Figure 4.3: Screenshot of the GECHO graphical user interface (GUI). Shown is an example of the selection of the integration boundaries for H$\beta$ and corresponding blue and red pseudo-continuum used to measure the integrated line flux.

The light curves of the continuum emission, that will be used as proxy for the ionizing continuum, are derived from the mean flux density of areas free from line emission. The continuum windows for the light curve extraction of the continuum emission are adopted from Cackett et al. 2018. The wavelength range for the continua are listed in Table 4.2.

Table 4.1: Integration Limits and Pseudo-Continua range of the measured emission-lines

| **Line** | **Integration Limits** [Å] | **Pseudo-Continua** [Å] |
|---|---|---|
| Ly$\alpha$ | $1207 - 1238$ | $1151 - 1161, 1461 - 1469$ |
| N V $\lambda\lambda$1238, 1242 | $1207 - 1238$ | $1151 - 1161, 1461 - 1468$ |
| Si IV $\lambda\lambda$1393, 1402 | $1358 - 1423$ | $1151 - 1161, 1461 - 1469$ |
| CIV $\lambda\lambda$1548, 1550 | $1511 - 1578$ | $1461 - 1469, 1680 - 1685$ |
| He II $\lambda$1640 | $1599 - 1645$ | $1461 - 1468, 1680 - 1685$ |
| H$\alpha$ | $6453 - 6695$ | $6107 - 6129, 6861 - 6900$ |
| H$\beta$ | $4779 - 4944$ | $4762 - 4774, 5085 - 5112$ |
| H$\gamma$ | $4230 - 4427$ | $4197 - 4220, 4435 - 4450$ |
| H$\delta$ | $4035 - 4165$ | $4026 - 4033, 4197 - 4220$ |
| He I $\lambda$5875 | $5742 - 6039$ | $5645 - 5653, 6044 - 6057$ |
| He II $\lambda$4685 | $4545 - 4758$ | $4435 - 4450, 4762 - 4774$ |
| O I $\lambda$8446 | $8380 - 8500$ | $8005 - 8031, 8850 - 8955$ |
| O III $\lambda$5007 | $4982 - 5033$ | $4762 - 4774, 5085 - 5112$ |

Table 4.2: Wavelength range of the measured continua

| **Line** | **Integration Limits** [Å] |
|---|---|
| Cont. 1150 | $1151 - 1161$ |
| Cont. 4010 | $4026 - 4033$ |
| Cont. 4440 | $4435 - 4450$ |
| Cont. 5100 | $5085 - 5112$ |
| Cont. 6110 | $6107 - 6129$ |
| Cont. 6880 | $6861 - 6900$ |
| Cont. 8015 | $8005 - 8031$ |
| Cont. 8900 | $8864 - 8955$ |

## 4.3 Light Curves

Thanks to the covered wavelength range, it is possible to measure several continuum light curves in the UV-, optical- and NIR regime. In addition, the photometric light curve of the *Swift* UVW2 band is included in the analysis, presented by McHardy et al. 2018. It as well has been used in Cackett et al. 2018 and exhibits a denser sampling than the HST/STIS light curves, with a cadence of 96 minutes for 6.4 d

between July 13 and July 18, 2016 and a central wavelength of about $\sim$ 1930 Å (McHardy et al. 2018).

All continuum light curves show a broadly similar shape (see Figure 4.4). The light curves start at high flux and then decrease by about 50–80% relative to their maxima within the first 1–4 days, reaching a minimum between days 4 and 5. Afterwards, the flux increases again in all light curves. The UV continua show a plateau-like behaviour from days 6 to 9, followed by a smaller decline between days 9 and 13, before rising again to a peak around day 13. A similar pattern is also noticeable in the other continuum light curves, but becomes less clear at higher wavelengths. While the optical continua also show higher flux levels between days 6 and 9 than between days 9 and 13, the near-IR continua exhibits a plateau over this interval. Towards the end of the campaign, the flux decreases again towards a minimum, with smaller short-term fluctuations, followed by a slight rise in the final days of the campaign in the UV and NIR bands.

The estimation of the physical distance between the SMBH and the region from which the emission-lines originate is the main goal of this reverberation mapping analysis. Assuming that the continuum radiation originates from the accretion disk, it is common to use the bluest available continuum as reference for the time lag estimation, which is expected to emerge closest to the SMBH (Peterson 1993). Therefore the *Swift* UVW2 continuum is used as a proxy light curve for the continuum's emission in this analysis.

Comparing the *Swift* UVW2 light curve to the measured broad emission-line light curves, they exhibit an equally strong decrease in flux, followed by a central part with higher flux and another decrease (see. Figure 4.5 and 4.6). By comparing these features, a correlation of the emission-line light curves to the *Swift* UVW2 light curve is already apparent with noticeable time lags which is further investigated in Section 4.4.

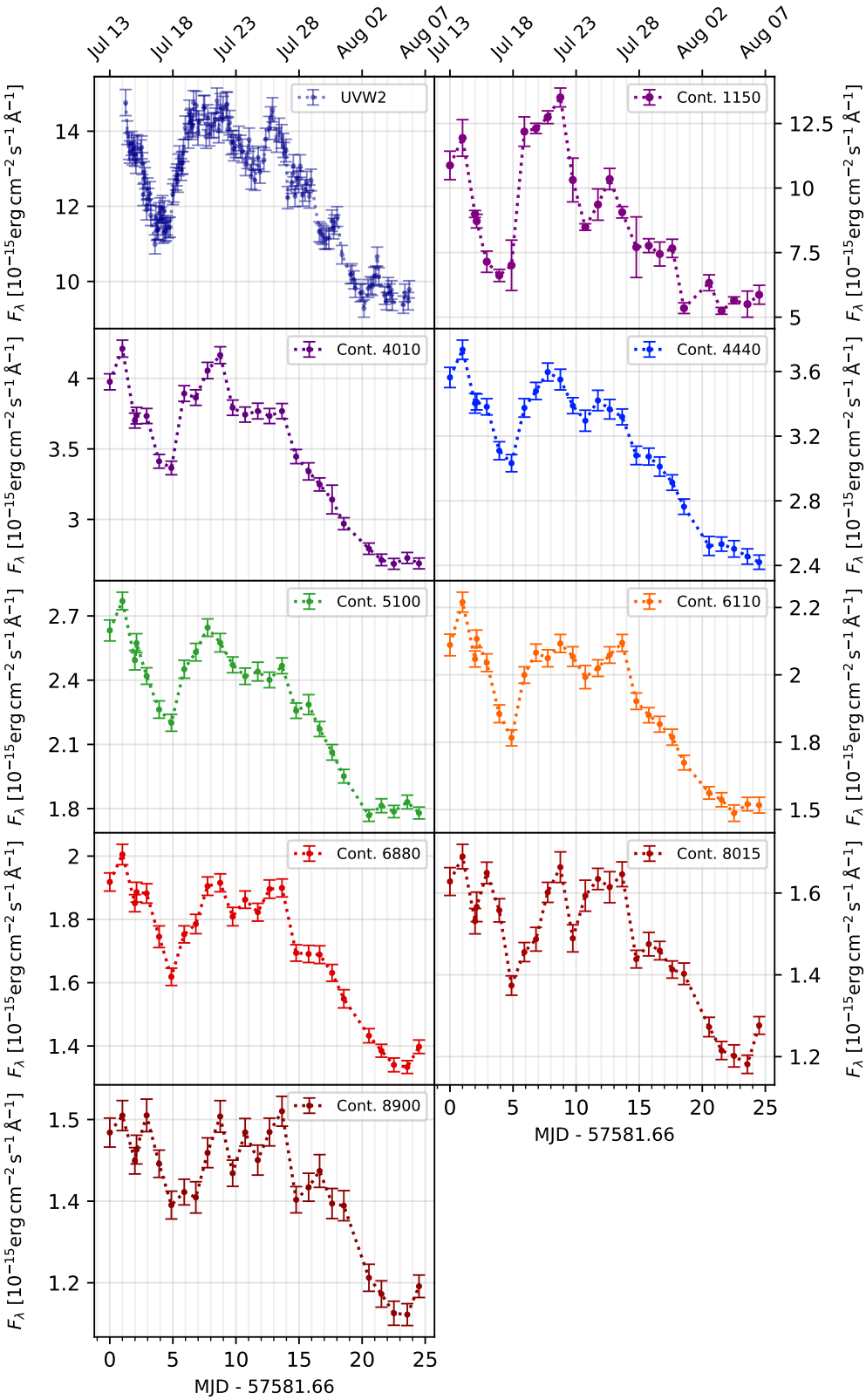


Figure 4.4: Comparison of the measured continua light curves of the HST/STIS dataset, together with the photometric *Swift* light curve of the UVW2 band obtained from McHardy et al. 2018, which is displayed in the first panel.

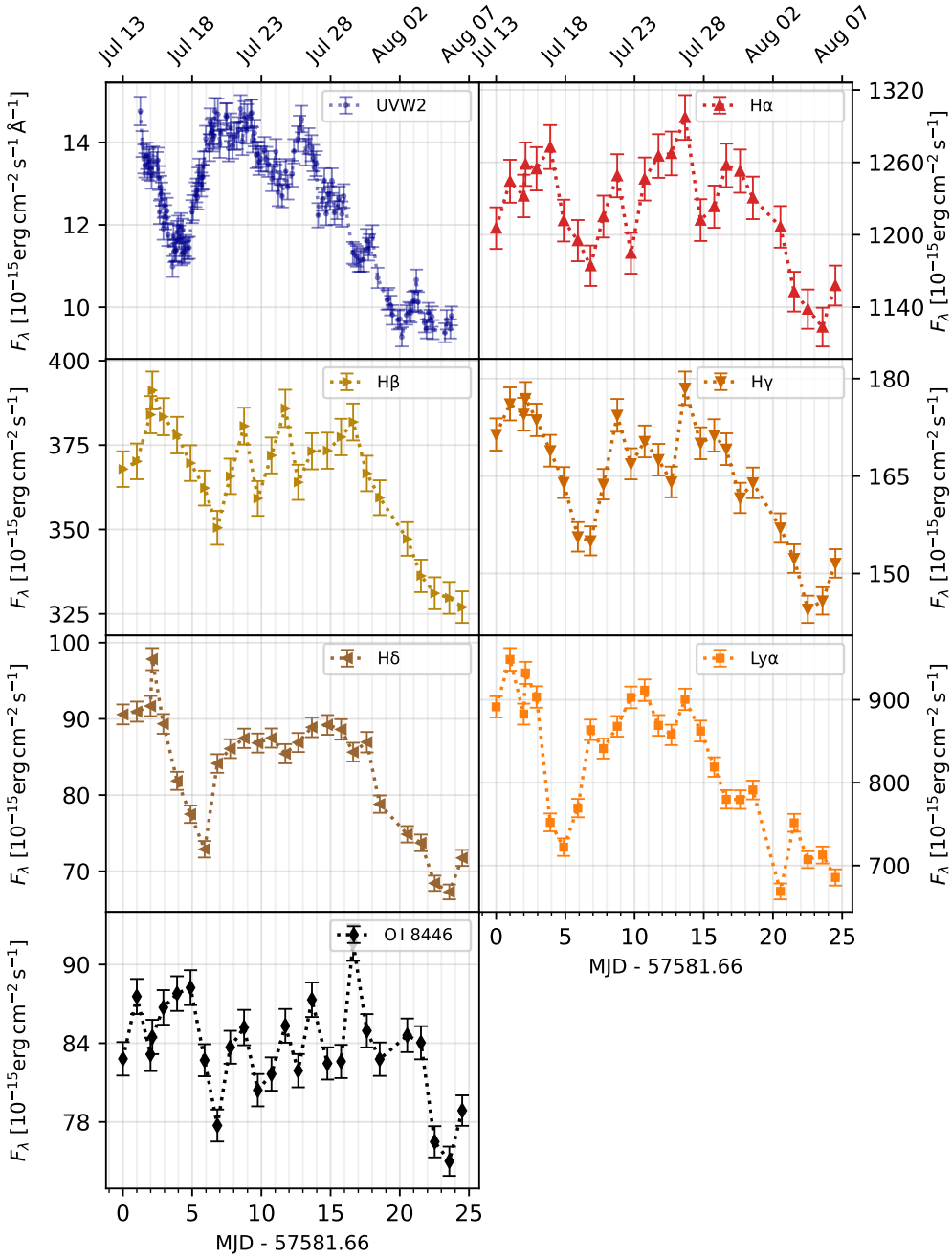


Figure 4.5: Comparison of the Balmer-line, Ly$\alpha$, and O I $\lambda$8446 light curves with the photometric *Swift* light curve of the UVW2 band as reference in the first panel. The *Swift* UVW2 light curve is adopted from McHardy et al. 2018.

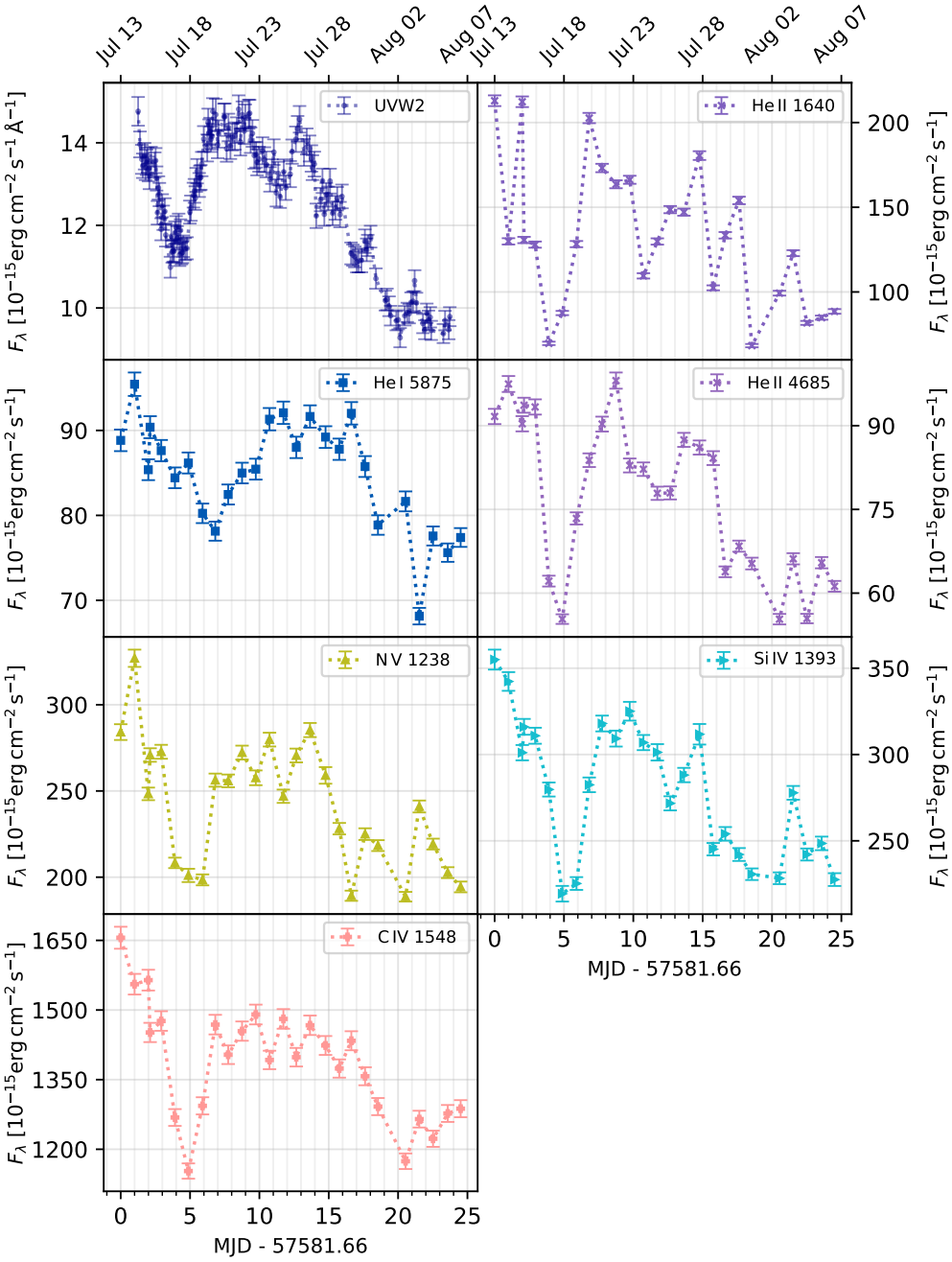


Figure 4.6: Comparison of the Helium and UV light curves with the photometric *Swift* UVW2 light curve as reference in the first panel. The *Swift* UVW2 light curve is adopted from McHardy et al. 2018.

### 4.3.1 Variability Statistics

The variability of the emission-lines and continua is quantified by adopting the variability statistics definition by Rodriguez-Pascual et al. 1997. They name the maximum-to-minimum flux ratio $R_{\rm max}$ and the fractional variability $F_{\rm var}$ as two common measures of variability. $R_{\rm max}$ is defined as the ratio of the extreme of the integrated fluxes, $F_{\rm min}$ and $F_{\rm max}$, and $F_{\rm var}$ as:

$$F_{\rm var} = \frac{\sqrt{\sigma_F^2 - \Delta^2}}{\langle F \rangle}. \tag{4.1}$$

Here, $\sigma_F^2$ denotes the standard deviation, $\langle F \rangle$ the mean flux and $\Delta^2$ the mean square value of the flux uncertainties, which is defined by:

$$\Delta^2 = \frac{1}{N} \sum_{i=1}^{N} \Delta_i^2 \tag{4.2}$$

The obtained $R_{\rm max}$ and $F_{\rm var}$ values of the continuum light curves indicates high variability in the UV regime with a decreasing trend towards the NIR regime (see Table 4.3). While the UV continuum around 1150 Å exhibits the highest values for $R_{\rm max} = 2.58$ and $F_{\rm var} = 0.28$, they continuously drop towards the most red continua around 8900 Å, which shows values of $R_{\rm max} = 1.33$ and $F_{\rm var} = 0.08$.
The Balmer emission-lines exhibit lower variability compared to the continua, which increases towards the higher-order Balmer lines, with values between $R_{\rm max} \simeq 1.15 - 1.52$ and $F_{\rm var} \simeq 0.03 - 0.1$. Together with H$\alpha$ and H$\beta$, the low-ionization OI $\lambda$8446 line exhibits the lowest variability of the measured emission-lines with $R_{\rm max} = 1.22$ and $F_{\rm var} = 0.04$. Ly$\alpha$ exhibits comparable variability towards the higher order Balmer lines with $R_{\rm max} = 1.42$ and $F_{\rm var} = 0.1$, as well as He I $\lambda$5876 with $R_{\rm max} = 1.62$ and $F_{\rm var} = 0.09$. The higher ionized He II $\lambda$1640 line shows a significant higher variability with $R_{\rm max} = 3.12$ and $F_{\rm var} = 0.31$, while He II $\lambda$4686 exhibits comparable values to He I $\lambda$5876 with $R_{\rm max} = 1.48$ and $F_{\rm var} = 0.11$. The high ionization lines tend to higher variability with $R_{\rm max}$ between $1.44 - 1.73$ and $F_{\rm var}$ between $0.9 - 0.15$. It shows that the higher ionized emission-lines tend to higher variability with the exception of He II $\lambda$4686 (see Table 4.4). A possible explanation for its lower fractional variability compared to He II $\lambda$1640 could be the blend with the Fe II band emission between approximately 4489 Å – 4629 Å, which adds a non variable component to the flux measurement of the light curve.

Table 4.3: Variability statistics of the measured continua with minimum flux $F_{\mathrm{min}}$ and maximum flux density or integrated flux $F_{\mathrm{max}}$, peak-to-peak ratio $R_{\mathrm{max}}$, mean $\langle F \rangle$, standard deviation $\sigma_F$ and fractional variation $F_{\mathrm{var}}$.

| **Continuum/Line** | $F_{\mathrm{min}}$ | $F_{\mathrm{max}}$ | $R_{\mathrm{max}}$ | $\langle F \rangle$ | $\sigma_F$ | $F_{\mathrm{var}}$ |
|---|---|---|---|---|---|---|
| Cont. 1150 | 0.52 | 1.35 | 2.58 | 0.86 | 0.25 | 0.28 |
| Cont. 4010 | 2.68 | 4.21 | 1.57 | 3.49 | 0.47 | 0.14 |
| Cont. 4440 | 2.42 | 3.73 | 1.54 | 3.14 | 0.39 | 0.12 |
| Cont. 5100 | 1.77 | 2.77 | 1.57 | 2.29 | 0.3 | 0.13 |
| Cont. 6110 | 1.49 | 2.27 | 1.53 | 1.9 | 0.23 | 0.12 |
| Cont. 6880 | 1.33 | 2.01 | 1.5 | 1.72 | 0.2 | 0.11 |
| Cont. 8015 | 1.18 | 1.69 | 1.43 | 1.48 | 0.15 | 0.1 |
| Cont. 8900 | 1.14 | 1.52 | 1.33 | 1.38 | 0.11 | 0.08 |

Table 4.4: Variability statistics of the measured emission-lines with minimum flux $F_{\mathrm{min}}$ and maximum flux density or integrated flux $F_{\mathrm{max}}$, peak-to-peak ratio $R_{\mathrm{max}}$, mean $\langle F \rangle$, standard deviation $\sigma_F$ and fractional variation $F_{\mathrm{var}}$.

| **Continuum/Line** | $F_{\mathrm{min}}$ | $F_{\mathrm{max}}$ | $R_{\mathrm{max}}$ | $\langle F \rangle$ | $\sigma_F$ | $F_{\mathrm{var}}$ |
|---|---|---|---|---|---|---|
| H$\alpha$ | 130.83 | 149.85 | 1.15 | 141.35 | 4.83 | 0.03 |
| H$\beta$ | 38.29 | 45.4 | 1.19 | 42.32 | 1.94 | 0.05 |
| H$\gamma$ | 18.82 | 24.38 | 1.3 | 21.91 | 1.29 | 0.06 |
| H$\delta$ | 7.17 | 10.92 | 1.52 | 9.13 | 0.94 | 0.1 |
| Ly$\alpha$ | 66.87 | 94.88 | 1.42 | 82.21 | 8.03 | 0.1 |
| O I $\lambda$8446 | 7.47 | 9.13 | 1.22 | 8.32 | 0.37 | 0.04 |
| He I $\lambda$5876 | 8.53 | 13.82 | 1.62 | 11.58 | 1.01 | 0.09 |
| He II $\lambda$1640 | 6.83 | 21.29 | 3.12 | 13.29 | 4.13 | 0.31 |
| He II $\lambda$4686 | 11.93 | 17.7 | 1.48 | 14.82 | 1.6 | 0.11 |
| N V $\lambda\lambda$1238, 1242 | 18.87 | 32.7 | 1.73 | 24.23 | 3.54 | 0.15 |
| Si IV $\lambda\lambda$1393, 1402 | 21.93 | 35.5 | 1.62 | 27.92 | 3.88 | 0.14 |
| C IV $\lambda\lambda$1548, 1550 | 115.31 | 165.57 | 1.44 | 138.77 | 12.12 | 0.09 |

### 4.3.2 Uncertainty Estimation

The uncertainties of the continuum and emission-line light curves are estimated based on the spectral noise and a systematic uncertainty introduced by the intercalibration.
For the continuum light curves, the per-epoch noise is estimated as

$$\sigma_i^{\text{cont}} = \frac{\sigma_{f_i}}{\sqrt{N}}, \tag{4.3}$$

where $\sigma_{f_i}$ denotes the standard deviation of the flux density within the selected continuum window in epoch $i$, and $N$ is the number of pixels within that window.
For the emission-line light curves, the noise is estimated from the scatter in the interpolated pseudo-continuum, $\sigma_i^{\text{noise}}$, as

$$\sigma_i^{\text{line}} = \frac{\sigma_i^{\text{p.cont}}\,\Delta\lambda}{\sqrt{N}}, \tag{4.4}$$

where $\Delta\lambda$ denotes the integration range of the emission-line and $N$ is the number of pixels across the integration window.
As each epoch has been scaled to the flux of the narrow [O III] $\lambda$5007 line after the intercalibration, the systematic uncertainty introduced by the intercalibration is estimated from its fractional variability $F_{\text{var}}^{[\text{O III}]\,\lambda5007}$:

$$\sigma_i^{\text{cal.}} = F_{\text{var}}^{[\text{O III}]\,\lambda5007}\, f_i. \tag{4.5}$$

The total uncertainty of the measured flux $f_i$ at epoch $i$ is then given with

$$\sigma_i^{\text{tot,cont}} = \sqrt{\left(\sigma_i^{\text{cal.}}\right)^2 + \left(\sigma_i^{\text{cont}}\right)^2}, \tag{4.6}$$

for the continuum light curves and

$$\sigma_i^{\text{tot,line}} = \sqrt{\left(\sigma_i^{\text{cal.}}\right)^2 + \left(\sigma_i^{\text{line}}\right)^2}, \tag{4.7}$$

for the emission-line light curves.

## 4.4 Time Lag

To measure the time lag between the continuum emission and the broad emission-lines, the emission-line light curves are correlated with the photometric *Swift* UVW2 light curve presented in McHardy et al. 2018. This is done by determining the cross-

correlation function (CCF) and measuring its centroid of the top 20% of the CCF peak, as described in Section 2.2.2. This calculation is performed with GECHO, which applies the interpolated cross-correlation function (ICCF) method introduced by Gaskell & Peterson 1987. For this purpose, the light curves are normalized by subtracting the respective mean value from each data point. To increase the effective sampling and enable the correlation of two unevenly sampled light curves, flux values between data points are linearly interpolated for both light curves. The centroid uncertainties are estimated using the flux randomization/random subset selection (FR/RSS) method described by Peterson et al. 1998, with 10.000 iterations and a lag bin size of 0.5 days. The normalized *Swift* UVW2 light curve and the emission-lines, together with the corresponding CCF and centroid distribution, are displayed in Figures 4.7 and 4.8. In addition, the auto-correlation function of the UVW2 light curve and its CCF with the continuum light curve around 1150 Å are delivered and displayed in the top and bottom panels of the figures, respectively.

Overall, the CCFs of all measured emission-lines show strong correlations with peak maximum values between 0.7 and 0.9, except for O I 8446, whose CCF exhibits a maximum value of approximately $\sim 0.6$. Since the *Swift* UVW2 light curve lags behind the light curve of the 1150 Å continuum by approximately $\sim 0.5$ days, this offset is added to the measured emission-line time lags to estimate the characteristic BLR radii, which are listed in Table 4.5.

H$\alpha$ and H$\beta$ lag behind the *Swift* UVW2 light curve by $3.3^{+1.1}_{-0.6}$ and $2.3^{+1.0}_{-0.3}$ days, respectively, while H$\gamma$ and H$\delta$ show shorter lags of $1.7^{+0.2}_{-0.4}$ and $1.7^{+0.4}_{-0.4}$ days. Therefore the lower order Balmer lines H$\alpha$ and H$\beta$ show higher time lags than the higher ordered Balmer lines H$\gamma$ and H$\delta$. Ly$\alpha$ exhibits an even shorter time lag than the Balmer lines with $1.0^{+0.4}_{-0.2}$ days within uncertainties, which implies that the hydrogen originate from different distances towards the SMBH.

He I 5875 shows a lag of $3.3^{+0.7}_{-0.4}$ days, which is of the same order as the lags measured for H$\alpha$ and H$\beta$ within the uncertainties. Opposing to that, the two higher ionized He II 1640 and He II 4685 helium lines show much shorter lags with $0.3^{+0.6}_{-0.2}$ days and $0.7^{+0.1}_{-0.3}$ days, respectively. The high ionized UV emission-lines N V 1238, Si IV 1393 and C IV 1548 also show short time lags of about $\sim 1$ day compared to the lower ionized Balmer and He I lines. Therefore a tendency becomes apparent, that the higher ionized emission-lines typically show shorter time lags than the lower ionized emission-lines.

The low-ionization line O I 8446 exhibits the longest lag of the monitored emission-lines with $4.7^{+2.4}_{-1.4}$ days and will be further discussed in Section 4.7.

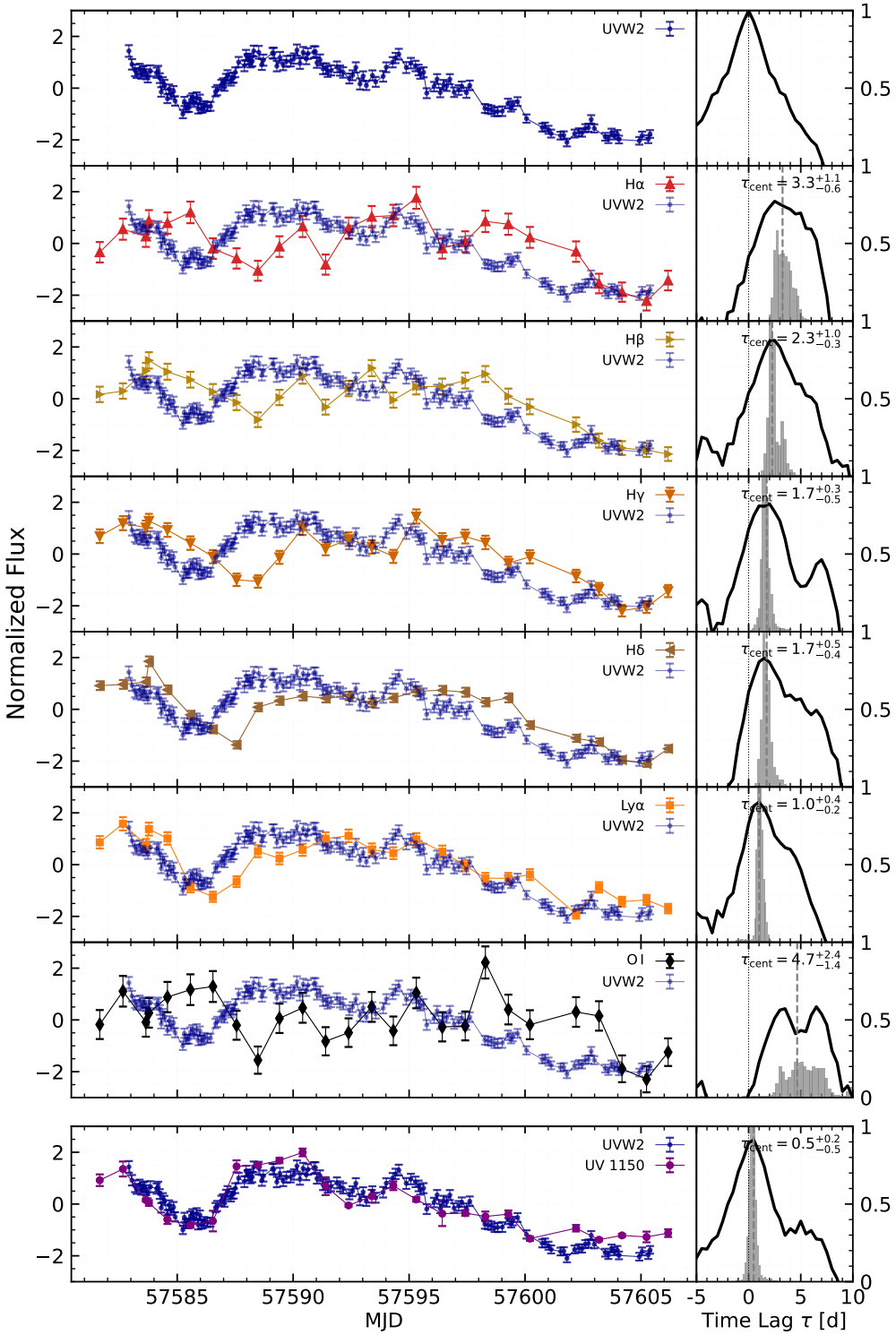


Figure 4.7: The left column displays a comparison of normalized light curves of the Balmer lines, He I $\lambda$5875, He II $\lambda$4685, Ly$\alpha$ and O I $\lambda$8446 with the photometric *Swift* UVW2 light curve, obtained from McHardy et al. 2018. The right column displays the corresponding CCF and centroid distribution. The first and last row of the figure display the auto-correlation function of the *Swift* UVW2 light curve and its CCF with the continuum light curve around 1150 Å, respectively.

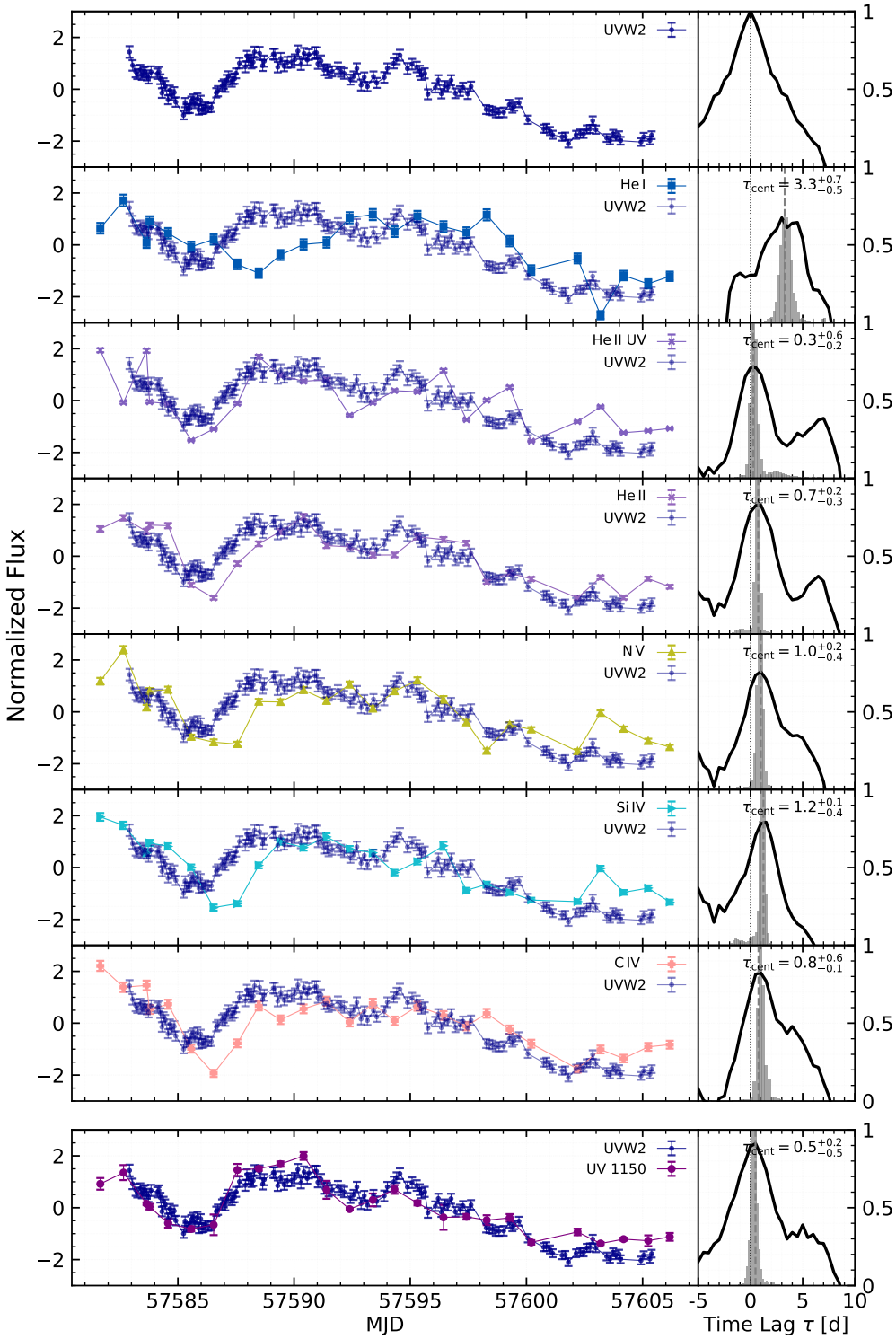


Figure 4.8: The left column displays a comparison of normalized light curves of the Helium lines and high-ionized UV lines with the photometric *Swift* UVW2 light curve, obtained from McHardy et al. 2018. The right column displays the corresponding CCF and centroid distribution. The first and last row of the figure display the auto-correlation function of the *Swift* UVW2 light curve and its CCF with the continuum light curve around 1150 Å, respectively.

Table 4.5: Centroid between the emission-lines and *Swift* UVW2 light curve and the characteristic BLR radii with an included shift of 0.5 light days due to the delay between *Swift* UVW2 and the UV continuum around 1150 Å.

| Line | $\tau_{\rm cent}$ [d] | $R_{\rm BLR}$ [ld] |
|---|---|---|
| H$\alpha$ | $3.3^{+1.1}_{-0.6}$ | $3.8^{+1.1}_{-0.6}$ |
| H$\beta$ | $2.3^{+1.0}_{-0.3}$ | $2.8^{+1.0}_{-0.3}$ |
| H$\gamma$ | $1.7^{+0.2}_{-0.4}$ | $2.2^{+0.2}_{-0.4}$ |
| H$\delta$ | $1.7^{+0.4}_{-0.4}$ | $2.2^{+0.4}_{-0.4}$ |
| Ly$\alpha$ | $1.0^{+0.4}_{-0.2}$ | $1.5^{+0.4}_{-0.2}$ |
| He I 5875 | $3.3^{+0.7}_{-0.4}$ | $3.8^{+0.7}_{-0.4}$ |
| He II 1640 | $0.3^{+0.6}_{-0.2}$ | $0.8^{+0.6}_{-0.2}$ |
| He II 4685 | $0.7^{+0.1}_{-0.3}$ | $1.2^{+0.1}_{-0.3}$ |
| N V $\lambda\lambda$1238, 1242 | $1.0^{+0.2}_{-0.4}$ | $1.5^{+0.2}_{-0.4}$ |
| Si IV $\lambda\lambda$1393, 1402 | $1.2^{+0.1}_{-0.4}$ | $1.7^{+0.1}_{-0.4}$ |
| C IV $\lambda\lambda$1548, 1550 | $0.8^{+0.5}_{-0.1}$ | $1.3^{+0.5}_{-0.1}$ |
| O I 8446 | $4.7^{+2.4}_{-1.4}$ | $5.2^{+2.4}_{-1.4}$ |

## 4.5 Line Profiles

The velocity dispersion of the BLR can be parameterized by the line widths of the broad emission-line (Peterson et al. 2004) and is common in reverberation mapping campaigns (e.g. Kollatschny & Dietrich 1997; Denney et al. 2006). Therefore the line profiles of the emission-lines are extracted from the mean and rms spectrum. Equally to the light curve extraction, a correction of the underlying continuum is necessary to enable comparison between the extracted profiles. Following the same procedure done for the light curves extraction, an underlying linear continuum is interpolated, using line-free wavelength windows on the blue and red side for each emission-line in both the mean and rms spectra. The selected boundaries of the pseudo-continua are listed in Table 4.6. By subtracting this interpolated continuum, a new zero-flux baseline for each line profile is defined. The flux is normalized to the maximum of each line profile and subsequently, the line profile is converted to velocity space using the relativistic Doppler equation (Sher 1968)

$$v_i = c \cdot \frac{\lambda_i^2 - \lambda_{\rm central}^2}{\lambda_i^2 + \lambda_{\rm central}^2}. \tag{4.8}$$

Here, $\lambda_i$ denotes the wavelength values, $\lambda_{\rm central}$ the central wavelength of the emission-line in rest-frame and $c$ the speed of light. For the doublets N V $\lambda\lambda$1283, 1242, Si IV $\lambda\lambda$1393, 1402 and C IV $\lambda\lambda$1548, 1550 the rest-frame wavelength of the second line gets used as the central wavelength of the emission-line. An overview of the

mean and rms profiles for each emission-line is displayed in Figure 4.9 and described in the following.

### Balmer lines

The mean profiles of the Balmer lines share an overall similar shape, characterized by a steep blue wing and an asymmetric broader red wing. The latter is not visible in the mean profile of H$\gamma$, since its red wing is blended with [O III] $\lambda$4363 emission. The rms profiles of the Balmer lines share an overall similar asymmetric double-peaked shape, with the blue peak exhibiting higher flux (see Figure 4.10). The two peaks are consistently located at approximately 500 km s$^{-1}$ and 2000 km s$^{-1}$ for all measured Balmer lines, resulting in a shift of the rms profiles towards positive velocities. While the profiles of H$\alpha$ and H$\beta$ show similarly steep slopes in both wings, those of H$\gamma$ and H$\delta$ display more extended blue wings, leading to a close resemblance between the rms profiles of H$\alpha$ and H$\beta$, and between H$\gamma$ and H$\delta$.

### Ly$\alpha$

The line profile of Ly$\alpha$ of both spectra exhibits absorption in the blue wing and is blended with N V $\lambda\lambda$1283, 1242 in the red wing. The rms profile showing a slightly broader line width than the mean profile and peak located at approximately 2000 km s$^{-1}$ and therefore as well shifted to positive velocities, like seen for the Balmer lines.

### O I $\lambda$8446

The mean profile of the NIR emission-line O I $\lambda$8446 is blended with the Ca II $\lambda$8498, $\lambda$8542, $\lambda$8662 triplet resulting in a asymmetric double-peak shape. In the rms spectrum, the profile is very noisy due to the low signal-to-noise ratio and exhibits therefore no pronounced shape.

### Helium Lines

The mean profile of He I $\lambda$5875 shows an asymmetric shape with broader red and steep blue wing. Its rms profile is shifted towards positive velocities, with the peak located at approximately $\sim$ 1500 km s$^{-1}$ and a line width comparable to that of the mean profile. The rms profile exhibits an asymmetric shape with a broader blue wing, though it is located mostly within the red wing of the mean profile due to the velocity shift. In contrast to He I $\lambda$5875, the mean profile of the higher ionized He II $\lambda$1640 line exhibits a asymmetric shape with a broader blue wing. Its

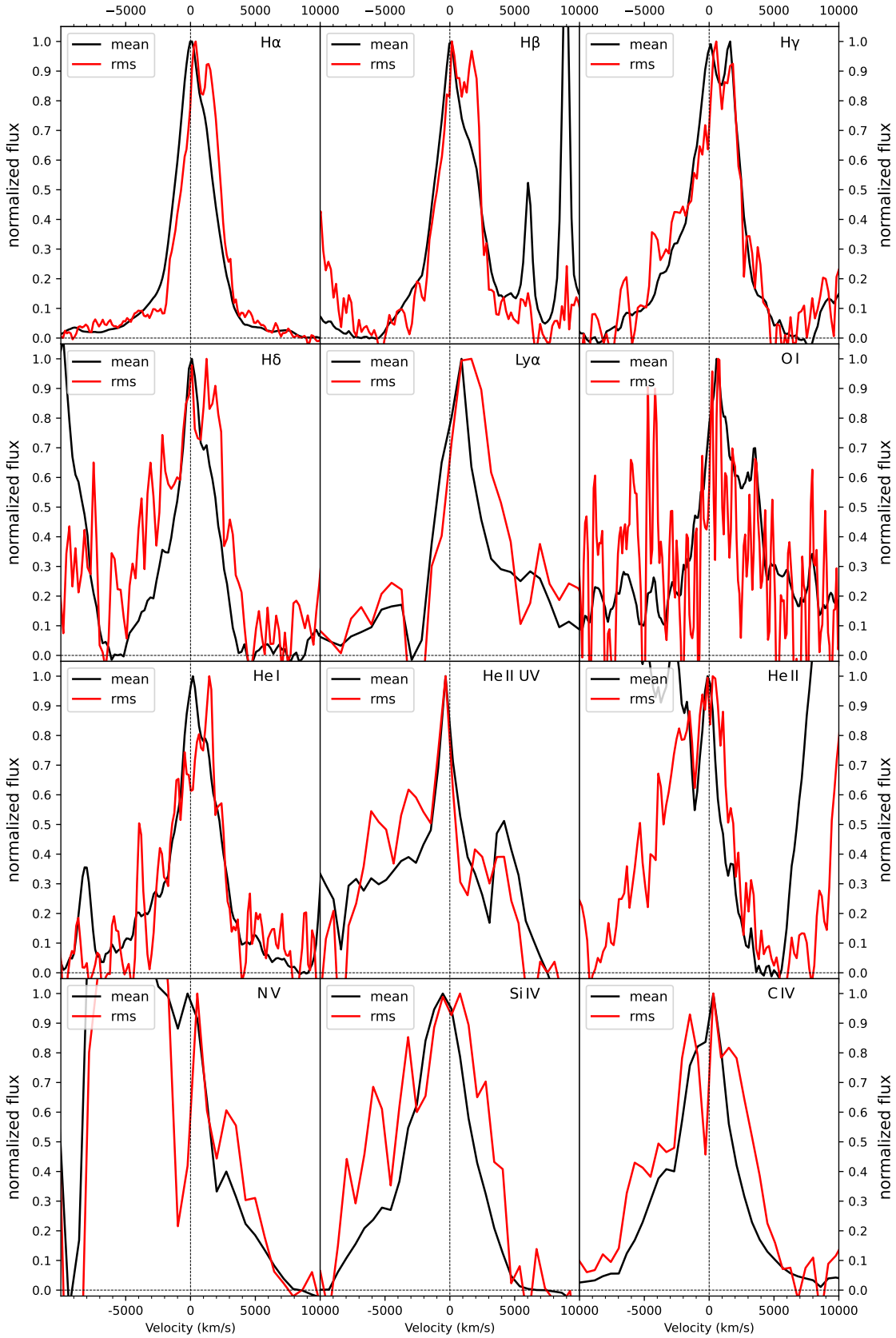


Figure 4.9: This figure displays the line profiles of the measured broad emission-lines in the mean spectrum (black) and rms spectrum (red).

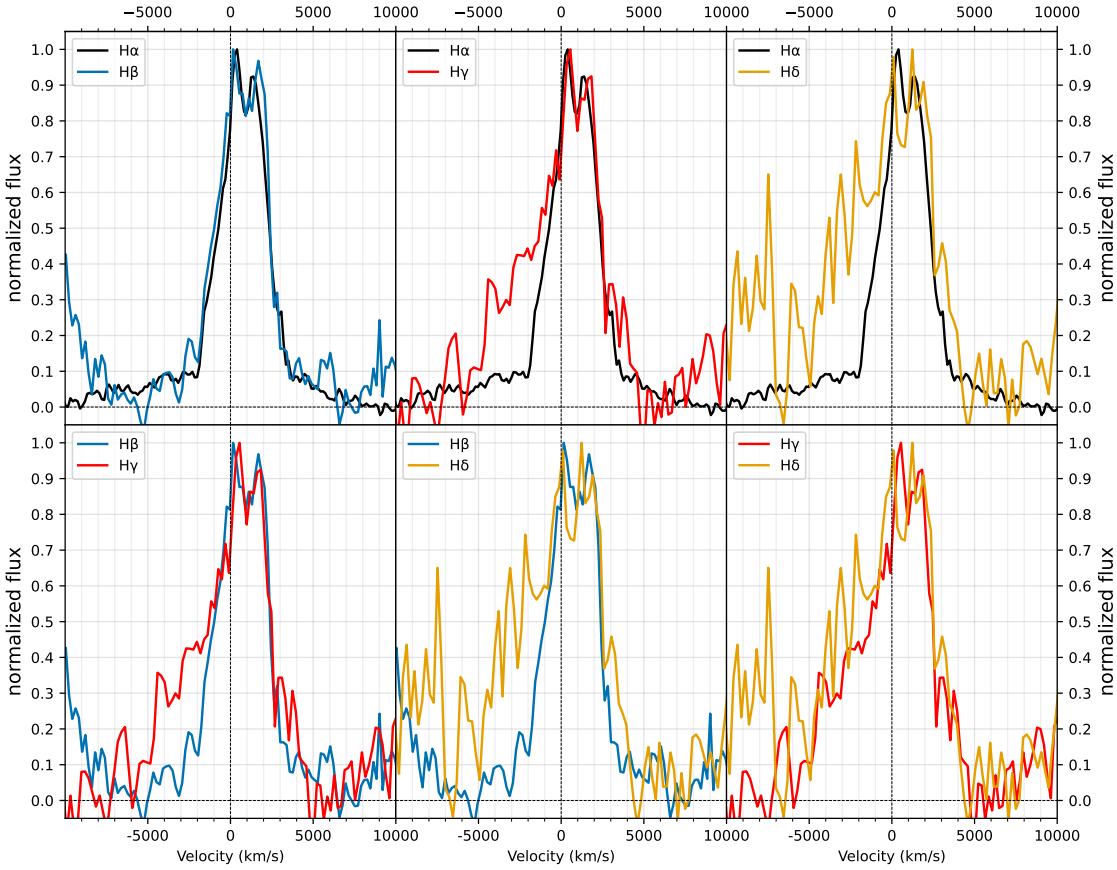


Figure 4.10: Comparison of the normalized rms line profiles of the Balmer lines H$\alpha$, H$\beta$, H$\gamma$ and H$\delta$.

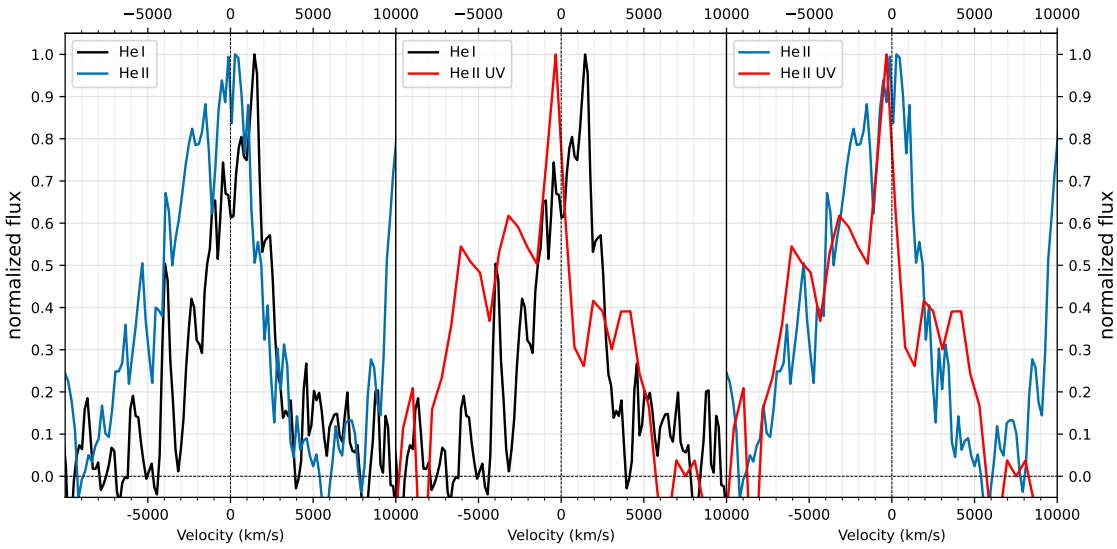


Figure 4.11: Comparison of the normalized rms line profiles of the Helium lines, He I $\lambda$5875, He II $\lambda$1640 and He II $\lambda$4685.

steeper red wing is blended with the semi-forbidden doublet O III]$\lambda\lambda$ 1660, 1666 in both the mean and rms spectra. The rms profile of He II $\lambda$4685 and He II $\lambda$1640 show a asymmetric shape with a extended blue wing like seen in the mean profile of He II $\lambda$1640. The blue wing of the mean profile of He II $\lambda$4685 is largely blended with Fe II emission.
While He I $\lambda$5875, He II $\lambda$1640 and He II $\lambda$4685 show overall comparable shapes when measured from their respective peaks, the He II lines exhibit significant broader line widths and no velocity shift relative to their mean profiles in contrast to He I $\lambda$5875 (see Figure 4.11).

**N V** $\lambda\lambda$1283, 1242

The N V $\lambda\lambda$1283, 1242 doublet emission is strongly blended with Ly$\alpha$, such that only the red wing of its mean profile is distinguishable from the Ly$\alpha$ mean profile. In addition, the doublet exhibits unresolved variable absorption in the rms spectrum, which further distorts the shape of the rms profile. Comparing the peak locations of both profiles, the rms profile shows no significant velocity shift, like it is seen for the He II lines.

**Si IV** $\lambda\lambda$1393, 1402

The mean profile of the Si IV $\lambda\lambda$1393, 1402 doublet shows an asymmetric shape with a more extended blue wing. It is blended with the semi-forbidden doublet OIV] $\lambda\lambda$ 1397, 1400, which cannot be resolved from the line profile. Similar to the mean profile, the rms profile exhibits a very broad blue wing, which additionally includes the variable component of the OIV] $\lambda\lambda$ 1397, 1400 doublet. The rms profile exhibits no significant velocity shift compared to its mean profile.

**C IV** $\lambda\lambda$1548, 1550

The mean profiles of the CIV $\lambda\lambda$1548, 1550 doublet shows an asymmetric shape with a broader blue wing, comparable to that of Si IV $\lambda\lambda$1393, 1402. The rms profile shows an even more extended blue wing and a double-peaked structure, caused by absorption near the center of the profile. As seen for the other high-ionization lines, the rms profile shows no significant velocity shift relative to the mean profile.

Comparing the mean and rms profiles, a clear distinction between lower- and higher-ionization emission-lines is noticeable. The rms profile peaks of the Balmer lines, Ly$\alpha$ and He I $\lambda$5875 are shifted towards positive velocities relative to the corre-

sponding mean profiles, with most of the rms emission located within the red wing of the mean profile. This indicates that the variability of these lines originates from the receding part of their emitting region. In contrast, the rms profile peaks of He II $\lambda$1640 and He II $\lambda$4685, as well as those of the high-ionization UV lines N V $\lambda\lambda$1283, 1242, Si IV $\lambda\lambda$1393, 1402 and C IV $\lambda\lambda$1548, 1550 show no velocity shift respective to their mean profiles. These higher-ionization lines show a blue-weighted rms profile, whereas the lower-ionization lines show a red-weighted rms profile. For N V $\lambda\lambda$1238, 1242, however, this behavior cannot be confirmed due to blending with the Ly$\alpha$ profile.

Table 4.6: Boundaries of the blue and red pseudo-continua used for the interpolation of underlying continua for line profile extraction.

| **Line** | **Pseudo-Continua (mean)** [Å] | **Pseudo-Continua (rms)** [Å] |
|---|---|---|
| H$\alpha$ | $6194-6216, 6861-6900$ | $6279-6301, 6742-6781$ |
| H$\beta$ | $4762-4774, 5085-5112$ | $4762-4774, 4967-4984$ |
| H$\gamma$ | $4197-4220, 4435-4450$ | $4197-4220, 4417-4429$ |
| H$\delta$ | $4026-4033, 4197-4221$ | $4006-4016, 4197-4211$ |
| O I $\lambda$8446 | $7999-8025, 8775-8798$ | $8222-8238, 8748-8767$ |
| He I $\lambda$5875 | $5679-5697, 6044-6057$ | $5736-5753, 6027-6045$ |
| He II $\lambda$1640 | $1461-1468, 1679-1685$ | $1461-1468, 1679-1685$ |
| He II $\lambda$4685 | $4198-4225, 4762-4774$ | $4543-4554, 4766-4778$ |
| Ly$\alpha$ | $1151-1161, 1270-1285$ | $1151-1161, 1340-1355$ |
| N V $\lambda\lambda$1283, 1242 | $1151-1161, 1270-1285$ | $1151-1161, 1340-1355$ |
| Si IV $\lambda\lambda$1393, 1402 | $1350-1360, 1430-1440$ | $1340-1355, 1430-1440$ |
| C IV $\lambda\lambda$1548, 1550 | $1461-1468, 1679-1685$ | $1461-1468, 1679-1685$ |

### 4.5.1 Full Width Half Maximum

Equally to earlier reverberation mapping studies (e.g. Kollatschny & Dietrich 1997; Denney et al. 2006; Probst et al. 2025), the line width of the line profiles is estimated by their Full Width Half Maximum (FWHM), which is conducted with GECHO. For this the height of the profiles is defined between the profiles maxima and the zero-flux base line. The mean profile of He II $\lambda$4685 is blended with Fe II emission in the blue wing, which makes a measurement of its FWHM not possible. Instead, the FWHM is estimated based on the width of its red wing, which is consistent with the order of magnitude of the FWHM of He II $\lambda$1640 within uncertainties. The same approach is used for N V $\lambda\lambda$1283, 1242, since it is blended with the Ly$\alpha$ profile in both mean and rms and exhibits not resolved absorption in its blue wing in the rms spectrum.

The FWHM of profiles of O I $\lambda$8446 and Si IV $\lambda\lambda$1393, 1402 were not measured. Since

the mean profile of the NIR emission-line O I $\lambda$8446 is blended with the Ca II $\lambda$8498, $\lambda$8542, $\lambda$8662 triplet and the rms profile shows a low signal-to-noise ratio, a FWHM measurement was not attempted. The profiles of Si IV $\lambda\lambda$1393, 1402 are blended with the semi-forbidden line doublet OIV] $\lambda\lambda$ 1397, 1400 such that any FWHM measurement would include contributions from both components and cannot be attributed to Si IV $\lambda\lambda$1393, 1402 alone, and is therefore excluded in the FWHM measurement. The resulting FWHM values are listed in Table 4.7. The uncertainties are estimated primarily from the instrumental dispersion of the respective gratings (see Table 3.1), and are adjusted according to the noise and shape of the profiles. For the FWHM of the mean profile of He II $\lambda$4685 and both profiles of N V $\lambda\lambda$1283, 1242 significantly larger uncertainties are adopted, as these values are estimated from only one wing of the profile.

The mean line profiles contain both broad and narrow components (Peterson 1997). Since the narrow component in the mean profiles can not be isolated, and the rms profiles contain only the variable broad component, the FWHM of the rms profiles are used in the subsequent SMBH mass estimation, and no decomposition of the mean profiles is attempted.

Comparing the FWHM of the rms profiles (see Table 4.7), the higher-order Balmer lines H$\gamma$ and H$\delta$ show larger FWHM than H$\alpha$ and H$\beta$, consistent with their more extended blue wings visible in Figure 4.10. The rms profile of Ly$\alpha$ exhibits an FWHM comparable to the higher order Balmer lines. Among the helium lines, a clear ionization dependence is apparent (see Figure 4.11) with the rms profiles of He II $\lambda$1640 and He II $\lambda$4686 show significantly broader FWHM than that of He I $\lambda$5876. The high-ionization line C IV $\lambda\lambda$1548, 1550 shows the largest FWHM among all measured broad emission-lines. The difference of the FWHM values between the higher and lower ionized helium lines as well as the significant higher FWHM of the high-ionization line in respect to the lower ionized Balmer lines therefore implies a ionization dependency of the rms profile line widths.

Table 4.7: Measured FWHM of the mean and rms line profiles.

| **Line** | FWHM (mean)[km/s] | FWHM (rms)[km/s] |
|---|---|---|
| H$\alpha$ | $3000 \pm 300$ | $3100 \pm 300$ |
| H$\beta$ | $3400 \pm 300$ | $3400 \pm 300$ |
| H$\gamma$ | $4100 \pm 300$ | $3900 \pm 400$ |
| H$\delta$ | $3400 \pm 300$ | $5900 \pm 2000$ |
| Ly$\alpha$ | $3800 \pm 350$ | $4600 \pm 500$ |
| He I $\lambda$5876 | $3500 \pm 300$ | $4000 \pm 400$ |
| He II $\lambda$1640 | $2900 \pm 1500$ | $6900 \pm 1500$ |
| He II $\lambda$4686 | $1800 \pm 1000$ | $6000 \pm 400$ |
| N V $\lambda\lambda$1238, 1242 | $3200 \pm 3000$ | $3400 \pm 2000$ |
| CIV $\lambda\lambda$1548, 1550 | $5400 \pm 500$ | $8400 \pm 1000$ |

## 4.6 Black Hole Mass

The mass of the SMBH is estimated based on the assumption that the velocity of the BLR is virialized and dominated by the gravitational potential of the SMBH (Peterson et al. 2004). Based on this assumption, a virial mass for each measured emission-line is determined with the found characteristic radii of the BLR and the line widths of the rms profiles, which only cover the virial component of the broad emission-line. The virial mass is then adjusting with a scale factor $f$ to account for unknown geometry, kinematic and inclination of the AGN (Peterson et al. 2004). Since the line widths of the rms are estimated with their FWHM, a scale factor of $f = 1.8$ is assumed, following Probst et al. 2025 (see Section 2.2.3). N V $\lambda\lambda$1238, 1242 is excluded in the mass estimation, due to the blend with Ly$\alpha$ and not resolvable absorption in its rms profile, which makes the found FWHM not reliable for the mass estimation.

Taking equation 2.7, 2.8 and 2.9, the SMBH mass can be estimated based on the time lag and FWHM for each emission-line using:

$$M_{\mathrm{BH}} = 1.8 \cdot \frac{c \cdot \tau_{\mathrm{centroid}}\,\mathrm{FWHM}^2}{G}\,. \tag{4.9}$$

The resulting SMBH mass results, as well as the corresponding time lag and FWHM values of the broad emission-lines are listed in Table 4.8.

To obtain a final estimation for the SMBH mass, an inverse-variance weighted mean is calculated. Since the uncertainties are asymmetric the higher gets adopted, $\sigma_i = \max(\sigma_i^-, \sigma_i^+)$. The weighted mean is then given by

$$\bar{M}_{\mathrm{BH}} = \frac{\sum_i w_i M_{\mathrm{BH},i}}{\sum_i w_i}\,, \qquad w_i = \frac{1}{\sigma_i^2}\,, \tag{4.10}$$

with an uncertainty

$$\sigma_{\bar{M}} = \left( \sum_i w_i \right)^{-1/2} . \tag{4.11}$$

This gives a weighted-mean SMBH mass of $\bar{M}_{\rm BH} \approx (1.40 \pm 0.12) \times 10^7\, M_\odot$.
It has to be noted that the scale factor of $f = 1.8$ does not account the low-inclination $i \sim 11°$ of the elliptic accretion disk, modeled in Ochmann et al. 2025. Adopting the scaling relation $f \sim \sin^{-2}(i)$ introduced by Krolik 2001, a value of $f \sim 27.5$ would be needed to account for the low inclination. Since Krolik 2001 used the line dispersion to parameterize the velocity dispersion, the relation $\sigma_{\rm line} \approx$ FWHM/2 (Peterson et al. 2004) must be applied again. Finally, due to the square relation of the velocity dispersion to the black hole mass seen in Equation 4.9, a scale factor of $f \sim 6.8$ is found. Subsequently the derived BH mass has to be multiplied by a additional factor of 3.77 resulting in a weighted-mean SMBH mass of $\bar{M}_{\rm BH} \approx (5.28 \pm 0.12) \times 10^7\, M_\odot$.

Table 4.8: Estimated characteristic BLR radii, FWHM and SMBH masses.

| **Line** | $R_{\rm BLR}$ [ld] | FWHM (rms)[km/s] | $M_{\rm BH}[10^7 M_\odot]$ |
|---|---|---|---|
| H$\alpha$ | $3.8^{+1.1}_{-0.6}$ | $3100 \pm 250$ | $1.3^{+0.4}_{-0.2}$ |
| H$\beta$ | $2.8^{+1.0}_{-0.3}$ | $3400 \pm 200$ | $1.2^{+0.5}_{-0.2}$ |
| H$\gamma$ | $2.2^{+0.2}_{-0.4}$ | $3900 \pm 300$ | $1.2^{+0.2}_{-0.3}$ |
| H$\delta$ | $2.2^{+0.4}_{-0.4}$ | $5900 \pm 1000$ | $1.9^{+0.4}_{-0.4}$ |
| Ly$\alpha$ | $1.5^{+0.4}_{-0.2}$ | $4600 \pm 500$ | $1.2^{+0.3}_{-0.2}$ |
| He I $\lambda$5876 | $3.8^{+0.7}_{-0.4}$ | $4000 \pm 300$ | $2.1^{+0.4}_{-0.3}$ |
| He II $\lambda$1640 | $0.8^{+0.6}_{-0.2}$ | $6900 \pm 1000$ | $1.3^{+1.0}_{-0.3}$ |
| He II $\lambda$4686 | $1.2^{+0.1}_{-0.3}$ | $6000 \pm 300$ | $1.6^{+0.2}_{-0.4}$ |
| C IV $\lambda\lambda$1548, 1550 | $1.3^{+0.5}_{-0.1}$ | $8400 \pm 1000$ | $3.2^{+1.3}_{-0.1}$ |

## 4.7 Variation and Bowen Fluorescence of O I λ8446

Emission-lines in the optical range, such as the Balmer and helium lines, are commonly used in reverberation mapping campaigns because they are readily accessible in nearby AGN (Peterson et al. 1991; Kollatschny & Dietrich 1997). This is not the case for the low-ionization line O I λ8446, which is of particular interest in this analysis because it exhibits noticeable variability (see Table 4.4) and can therefore be investigated with reverberation mapping for the first time. O I λ8446 is also known as a Bowen fluorescence line that can be pumped by Ly$\beta$ emission (Grandi 1980). As described in Section 2.3, Ly$\beta$ excites O I via a near-resonant transition at λ 1025 Å. This is followed by line transition through the O I λ 11287 transition and subsequently the O I λ 8446 transition, producing the observed O I λ 8446 emission

(see Figure 2.5). This process has been identified as the dominant excitation mechanism of O I $\lambda$8446 in many AGN, rather than recombination (e.g. Rudy et al. 1989; Rodriguez-Ardila et al. 2002; Landt et al. 2008). If recombination were a significant driver of the O,I $\lambda$8446 emission, the quintet line O I $\lambda$7774 would be expected to be comparatively strong, with an expected ratio of $\lambda7774/\lambda8446 \approx 1.1$–$1.7$ (Grandi 1980; Landt et al. 2008). While O I $\lambda$7774 is not detected in the HST dataset, a ratio of $\lambda7774/\lambda8446 = 0.2$ can be estimated from the MUSE spectrum of NGC 4593 presented by Ochmann et al. 2025. Therefore recombination can be excluded as a significant excitation mechanism for O I $\lambda$8446 in NGC 4593, which indicated that its variability of O I $\lambda$8446 is primarily driven by L$\beta$ Bowen fluorescence. This allows the Bowen-fluorescence relation of O I $\lambda$8446 to be examined in a reverberation mapping analysis for the first time.

While Ly$\beta$ is not covered in the spectral range of the HST dataset, it is possible to use Ly$\alpha$ as a proxy for Ly$\beta$, since it is expected that both lines arise under similar physical conditions. Therefore, the time lags between the *Swift* UVW2 and emission-line light curves, as well as the CCFs and time lags between O I $\lambda$8446 and Ly$\alpha$, between H$\alpha$ and Ly$\alpha$, and between O I $\lambda$8446 and H$\alpha$ are calculated and shown in Figure 4.12. All light curve show a strong correlation between $\sim 0.75 - 0.8$, with the exception of the O I $\lambda$8446 and *Swift* UVW2 light curves, which show a maximum correlation of only 0.6. The time lags and their uncertainties are derived as before. O I $\lambda$8446 lags behind Ly$\alpha$ by $2.5^{+1.7}_{-0.3}$ days and behind *Swift* UVW2 by $4.7^{+2.4}_{-1.4}$ days, while H$\alpha$ lags behind Ly$\alpha$ by $1.8^{+0.5}_{-0.5}$ days and behind *Swift* UVW2 by $3.3^{+1.1}_{-0.6}$ days. Therefore the direct relation of O I $\lambda$8446 to UVW2 shows a slightly higher lag than H$\alpha$ to UVW2 within uncertainties. Assuming that the Bowen fluorescence is the dominant excitation mechanism, the lag of O I $\lambda$8446 has to be estimated by adding the time lag between the Ly$\alpha$ light curve and the *Swift* UVW2 light curve and the time lag between the Ly$\alpha$ and O I $\lambda$8446 light curve. With the Ly$\alpha$ light curve lagging behind the *Swift* UVW2 light curve by $1.0^{+0.4}_{-0.2}$ days, this sums up to a time lag of approximately $\sim 3.5$ days, which is consistent with the time lag between the H$\alpha$ and *Swift* UVW2 light curve within uncertainties. Looking at the O I $\lambda$8446 and H$\alpha$ light curves, they show a strong correlation, with O I $\lambda$8446 lagging behind H$\alpha$ by about $0.3^{+1.6}_{-0.1}$ days. This implies that O I $\lambda$ 8446 is emitted at approximately the same distance as H$\alpha$ within uncertainties.

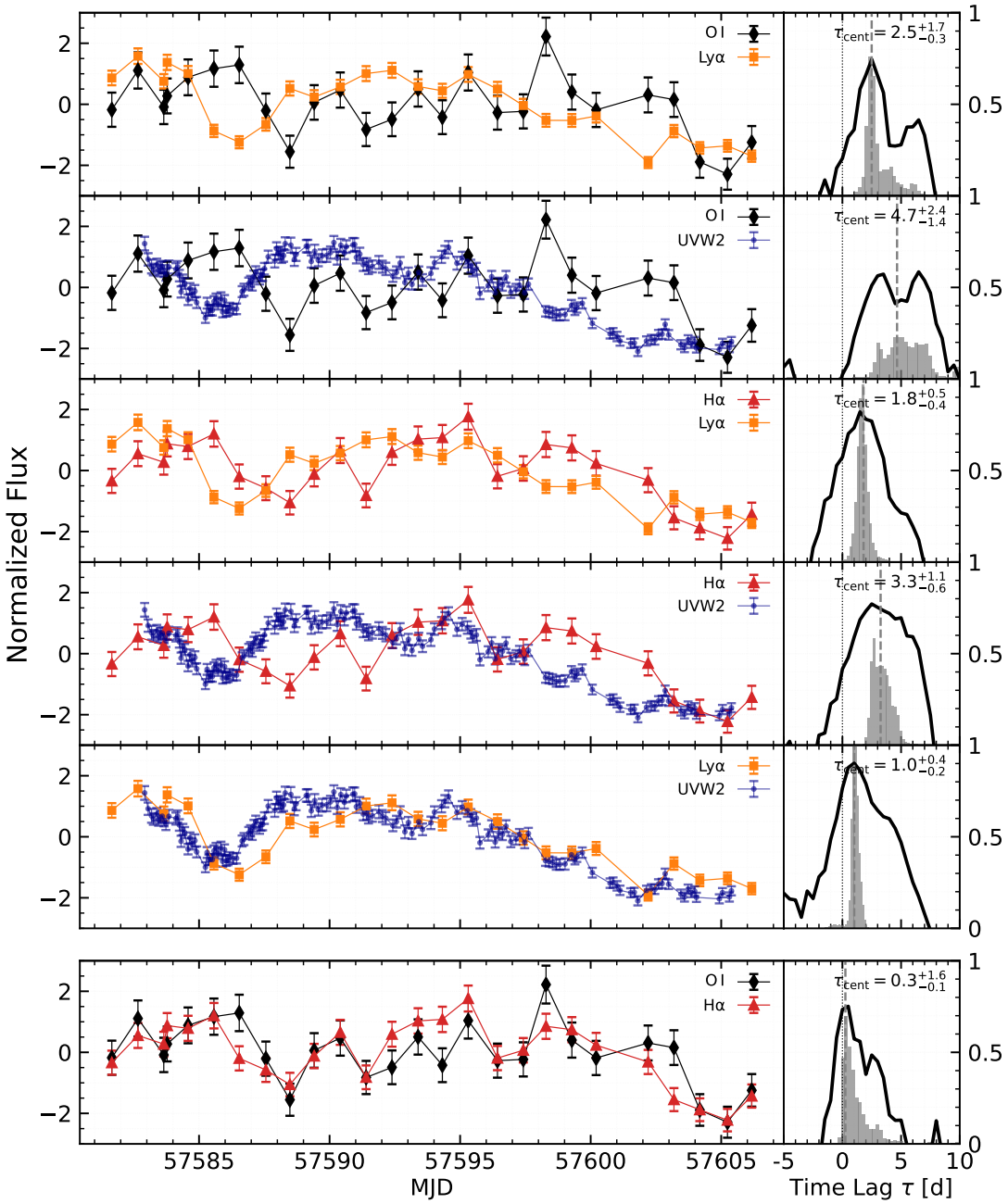


Figure 4.12: The left column displays a comparison of normalized light curves of O I vs Ly$\alpha$, O I vs *Swift* UVW2, H$\alpha$ vs Ly$\alpha$, H$\alpha$ vs *Swift* UVW2, Ly$\alpha$ vs *Swift* UVW2 and O I vs H$\alpha$ and their corresponding CCFs. The right column displays the corresponding CCF and centroid distribution.

# 5. Discussion

In a recent study, Cackett et al. 2018 conducted an observing campaign of NGC 4593 with HST/STIS (PI: Cackett, E.M, Prop ID 14121), covering the optical to NIR between approximately 2900 Å – 10000 Å as well as the UV range between approximately $\sim$ 1100 Å–1700 Å. The study focused on a reverberation mapping analysis of the UV and optical continua emission of NGC 4593 (Cackett et al. 2018). However, a reverberation mapping analysis of the BLR based on this dataset had not yet been attempted. Building on the dataset of Cackett et al. 2018, this thesis presents the first reverberation-mapping analysis of the BLR of NGC 4593 based on HST/STIS observations, including broad emission-lines that have not been monitored in a reverberation mapping campaign of NGC 4593 yet. In addition, variability in the low-ionization line O I $\lambda$ 8446 was measured in the HST/STIS dataset (see Table 4.4), which was never monitored in a reverberation-mapping campaign before. This enabled the first reverberation mapping analysis of the O I $\lambda$ 8446 line.

## 5.1 Structure of the BLR

### 5.1.1 Time Lags

The extracted light curves of the emission-lines show an overall similar shape, with strong correlation with the *Swift* UVW2 light curve and CCF peak values between 0.7–0.9. Only the O I $\lambda$ 8446 light curve show a lower peak correlation with *Swift* UVW2 of 0.6, which is consistent with the exclusion of recombination as significant excitation mechanism for O I $\lambda$ 8446 (see Section 4.7).
H$\alpha$ and H$\beta$ show different time lags of $3.3^{+1.1}_{-0.6}$ and $2.3^{+1.0}_{-0.3}$ days which are in good agreement with the results of earlier works (Kollatschny & Dietrich 1997; Denney et al. 2006; Williams et al. 2018). The higher-order Balmer lines H$\gamma$ and H$\delta$ show a significantly shorter time lags with $1.7^{+0.2}_{-0.4}$ days and $1.7^{+0.4}_{-0.4}$ days, respectively. This implies radially stratified structure of the Balmer-line emitting region, where higher-order lines respond more rapidly to continuum variations than lower-order

lines, consistent with optical depth effects observed for the Balmer-line emitting region (Korista & Goad 2004)
Looking at the helium lines, they show different time lags depending on their ionization. He I $\lambda$5876 lags $3.3^{+0.7}_{-0.4}$ days behind *Swift* UVW2, while the higher-ionized He II $\lambda$1640 and He II $\lambda$4686 exhibit a much faster response time, with lags of $0.3^{+0.6}_{-0.2}$ days and $0.7^{+0.1}_{-0.3}$ days, respectively. Therefore the time lag of He I $\lambda$5876 is consistent with the lag of H$\alpha$, while the He II lines exhibits significantly shorter delays.
The high-ionized UV lines all show similar lags within uncertainties, with measured values of $1.0^{+0.2}_{-0.4}$, $1.2^{+0.1}_{-0.4}$ and $0.8^{+0.5}_{-0.1}$ days for N V $\lambda\lambda$1238, 1242, Si IV $\lambda\lambda$1393, 1402 and CIV $\lambda\lambda$1548, 1550, respectively.
The ionization dependency seen for the helium line time lags and the consistent smaller time lags of high-ionization lines compared to the the lower ionized Balmer lines implies an ionization stratification structure of the BLR. This is consistent with BLR structures reported for other Seyfert galaxies (e.g Collin-Souffrin & Lasota 1988; Gaskell & Sparke 1986; Kollatschny et al. 2001).
Since the UVW2 band lags behind the extracted HST/STIS 1150 Å continuum by approximately 0.5 days, this offset was added for the final estimation of the characteristic BLR radii for each emission-line (see Table 4.5).

### 5.1.2 Line Profiles

Based on the assumption that the velocity of the BLR is virialized and dominated by the gravitational potential of the SMBH, the velocity dispersion of the BLR can be parameterized by the line widths of the rms line profiles broad emission-line (Peterson et al. 2004) and is common in reverberation mapping campaigns(e.g. Kollatschny & Dietrich 1997; Denney et al. 2006)
The Balmer rms line profiles show a similar asymmetric, double-peaked structure and are shifted toward positive velocities. This shift is also noticeable in the Ly$\alpha$ profile and He I $\lambda$5876 profile. This indicates that most of the line variability of these lines originates from the receding part of the emitting region.
Opposing to that, the maxima of the higher-ionization He II $\lambda$1640 and He II $\lambda$4686 profiles show no velocity shift. A similar behaviour is observed in the rms profiles of the high-ionized UV lines N V $\lambda\lambda$1238, 1242, Si IV $\lambda\lambda$1393, 1402 and CIV $\lambda\lambda$1548, 1550.
High-ionized broad emission-lines often show a blue-shift compared to lower-ionized broad emission-lines (Gaskell 1982), which is a possible explanation for the shift in the rms profile.
The measured FWHM values of the Balmer rms line profiles agree with a radial stratification structure of the emitting region of the Balmer lines (Korista & Goad

2004). While H$\alpha$ and H$\beta$ exhibits FWHM of $(3100 \pm 300)$ and $(3400 \pm 300)$, H$\gamma$ and H$\delta$ exhibits broader FWHM values of $(3900 \pm 400)\,\mathrm{km\,s^{-1}}$ and $(5900 \pm 2000)\,\mathrm{km\,s^{-1}}$, respectively.
The FWHM of the rms profiles of show a clear ionization dependence. The higher ionized helium lines He II $\lambda$1640 and He II $\lambda$4686 show significant broader line widths than the lower ionized helium line He I $\lambda$5876 with $(6900 \pm 1000)\,\mathrm{km\,s^{-1}}$, $(6000 \pm 400)\,\mathrm{km\,s^{-1}}$ and $(4000 \pm 400)\,\mathrm{km\,s^{-1}}$ for He I $\lambda$5876, respectively. The high-ionization line doublet C IV $\lambda\lambda$1548, 1550 exhibits the broadest rms profile with $(8400 \pm 500)\,\mathrm{km\,s^{-1}}$, which is significantly broader compared to the lower ionized emission-lines like the Balmer lines. Therefor the measured FWHM are consisted with the inferred ionization stratification structure of the estimated characteristic BLR radii.

### 5.1.3 Variability

The variability of the measured continuum and broad line emission was quantified with the fractional variability, as defined in Rodriguez-Pascual et al. 1997. The measured fractional variability of the continuum light curves follows the same trend as reported in Cackett et al. 2018, with the highest variability in the UV and decreasing variability toward the NIR.
Similar to the time lags and line widths, the Balmer line light curves show also show a variability trend, with the highest fractional variability in H$\delta$ and the lowest in H$\alpha$, which is consistent with a radial stratification structure of the Balmer region.
Also, the emission-line light curves overall show higher variability for more highly ionized lines, with the exception for the He II $\lambda$4686 light curve. A possible explanation for this discrepancy may be the blend with non variable Fe II emission interfering with the flux measurement of the line. Therefore, the observed variability trend of the emission-lines is as well consistent with an ionization stratification structure of the BLR with more highly ionized lines tend to show higher variability.

## 5.2 Black Hole Mass

Based on the reverberation mapping results, virial masses were derived for the selected emission-lines, and the SMBH mass was estimated using a scale factor of $f = 6.9$. The adopted scale factor accounts for the low inclination angle of $i \sim 11°$ reported by Ochmann et al. 2025. Its estimation is based on the relation $f \sim \sin^{-2}(i)$ (Krolik 2001), which captures the inclination dependence of the observed line-of-sight velocity dispersion. Since in this work, the velocity width is parameterized via the FWHM rather than the line dispersion, an additional conversion was applied

using $\sigma_{\rm line} \approx$ FWHM/2 (Peterson et al. 2004). Using this approach, an inverse-variance–weighted mean BH mass of $\bar{M}_{\rm BH} \approx (5.28 \pm 0.12) \times 10^7\,M_\odot$ was obtained from the determined values of each suitable emission-line. Earlier works reported values in the same order of magnitude, e.g. $M \approx 1.4 \times 10^7\,M_\odot$ (Kollatschny & Dietrich 1997) and $M = (9.8 \pm 2.1) \times 10^6\,M_\odot$ (Denney et al. 2006). Using a similar approach, Kollatschny & Dietrich 1997 monitored H$\alpha$ and estimated the SMBH mass using a scale factor of 1.5, whereas Denney et al. 2006 monitored H$\beta$ and used a scale factor of 5.5. Rescaling these values to the scale factor of 6.9 used here yields $M \approx 6.44 \times 10^7\,M_\odot$ (Kollatschny & Dietrich 1997) and $M = (1.23 \pm 0.26) \times 10^7\,M_\odot$ (Denney et al. 2006). Therefore, the SMBH mass found in this analysis is consistent in order of magnitude with these earlier estimates.

## 5.3 Variation and Bowen Fluorescence of O I $\lambda$8446

The low-ionization line O I $\lambda$ 8446 exhibits a noticeable variability with $F_{\rm var} = 0.04$ in the HST/STIS dataset. Since O I $\lambda$ 8446 was never monitored within a reverberation-mapping campaign before, this work presents the first results of a reverberation-mapping analysis for this line. In addition to that, O I $\lambda$ 8446 is known to be a Bowen fluorescence line, which can be enhanced by Ly$\beta$ emission due to a near coincidental transition (Grandi 1980). Due to the absence of quintet emission-lines, recombination was ruled out as a significant excitation mechanism for O I $\lambda$ 8446 (Grandi 1980). In addition it showed, that O I $\lambda$ 8446 exhibits a significantly stronger correlation with Ly$\alpha$ than with UVW2. Ly$\beta$ excites O I through a near-resonant transition at $\lambda$ 1025 Å, followed by line transition via the O I $\lambda$ 11287 transition and subsequently by the O I $\lambda$ 8446 transition (see Figure 2.5). This cascade is expected to occur with a photon ratio of 1:1 (Grandi 1980). A photon ratio of $\sim 0.8$ between O I $\lambda$ 8446 and O I $\lambda$ 11287, which was taken from the near-infrared SpeX spectra obtained by Landt et al. 2008, was determined. This result shows that Ly$\beta$ pumping is the dominant excitation mechanism in O I $\lambda$ 8446 in NGC 4593. Therefore it can be assumed that the O I $\lambda$8446 variation in NGC 4593 is driven through Bowen fluorescence, which enables to examine the Bowen fluorescence relation of O I $\lambda$8446 for the first time in a reverberation-mapping analysis. Since Ly$\beta$ is not covered in the HST/STIS dataset, Ly$\alpha$ was used as a proxy for investigation of the Bowen fluorescence O I $\lambda$ 8446. Therefore, the emitting region of O I is likely located at a distance corresponding to a lag of $\sim 3.5$ days from the ionizing continuum, consistent with the Bowen fluorescence scenario, rather than the $4.7^{+2.4}_{-1.4}$ days implied by the direct lag relative to UVW2 under the assumption of recombination as the main

excitation mechanism. With that, the Bowen fluorescence lag of O I corresponds to the radius of the emitting region of H$\alpha$ with $3.3^{+1.1}_{-0.6}$ days. Since H$\alpha$ and O I $\lambda$ 8446 also exhibits a strong correlation with a minimal time shift of $\sim$ 0.3 days, it can be inferred that O I $\lambda$ 8446 is emitted at approximately the same characteristic BLR radius as H$\alpha$ within uncertainties.

## 5.4 Summary

This thesis presents the results of the first reverberation-mapping analysis of the broad-line region (BLR) of the Seyfert I galaxy NGC 4593 based on an HST/STIS campaign (PI: Cackett, E.M, Prop ID 14121). This dataset is ideally suited for reverberation mapping thanks to its daily cadence and its broad wavelength coverage spanning parts of the UV and the optical-to-NIR regimes. This enabled a more detailed investigation of the BLR structure than in previous reverberation mapping studies of NGC 4593 by including additional broad emission-lines that had not been covered before. The measured time lags and FWHM values are listed in Tables 4.5 and 4.7, respectively. The results indicate a systematic trend: higher-ionization broad emission-lines exhibit shorter time lags and broader line widths than lower-ionization lines, indicating an ionization stratified BLR (Collin-Souffrin & Lasota 1988).

Based on the virial products of the reverberation-mapped lines, an SMBH mass of $\bar{M}_{\rm BH} \approx (5.28 \pm 0.12) \times 10^7\, M_\odot$ was inferred, which agrees in order of magnitude with earlier estimates (e.g. Kollatschny & Dietrich 1997; Denney et al. 2006).

In addition, variability of the low-ionization line O I $\lambda$ 8446 was detected for the first time in a reverberation-mapping analysis. It was shown that this variability is primarily driven through Ly$\beta$ fluorescence, making O I $\lambda$ 8446 the first Bowen-fluorescence line investigated with reverberation mapping. Following the Bowen-fluorescence process, the emitting region was approximately found at the same distance as the emitting region of H$\alpha$ towards the ionizing continuum. The results of the analysis of the O I $\lambda$ 8446 variability further led to the publication (Ochmann et al. 2026).

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